

Cairn

How to use the Cairn touch interface

Messaging, nodes, radio tools, GPS and maps, settings, power and updates — everything on the device, explained screen by screen.



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Screens are drawn at the panel's 320 × 240 resolution with fictional sample data. Cairn is an independent project and is not affiliated with Seeed Studio. MeshCore is MIT-licensed, Copyright (c) 2025 Scott Powell / rippleradios.com.

01 Welcome

Cairn turns the Seeed Wio Tracker L2 Pro into a stand-alone LoRa mesh messenger: you read, reply, find nodes, watch the radio and get a GPS fix on the device itself, without a phone. It is a build of the MeshCore companion firmware (v1.17.1) with a native touch interface, so it still pairs with the official MeshCore apps over Bluetooth whenever you want a bigger screen.

The manual follows the device from the outside in: first how to install Cairn and get through the first run, then the parts of the interface you use everywhere (the icon rail, the status bar, the buttons, the lock screen and the Quick panel), then one chapter per page of the interface, and finally power, updates and a reference section.

The screen images are drawn at the panel's real 320 x 240 resolution and use sample data: the names, channels and positions are fictional.

02 Getting started

A first install takes one USB flash of the full image; after that, updates can come over USB or Wi-Fi.

Install Cairn over USB

1. Connect the L2 Pro to your computer with a USB-C data cable.
2. Put the board in download mode: hold **User/Boot**, tap **RST**, then release **User/Boot**.
3. Write the full image at address 0x0:

```
esptool.py --chip esp32s3 write_flash 0x0 MeshCore-WioL2Pro-Cairn105-<date>-full.bin
```

4. Tap **RST** to start Cairn. A browser-based ESP32 flasher works too, with the same full image at 0x0.

Later builds can be installed with the smaller `-update.bin` written at 0x10000, or without a cable through Settings → System → Firmware update (see Updating and recovery).

First run

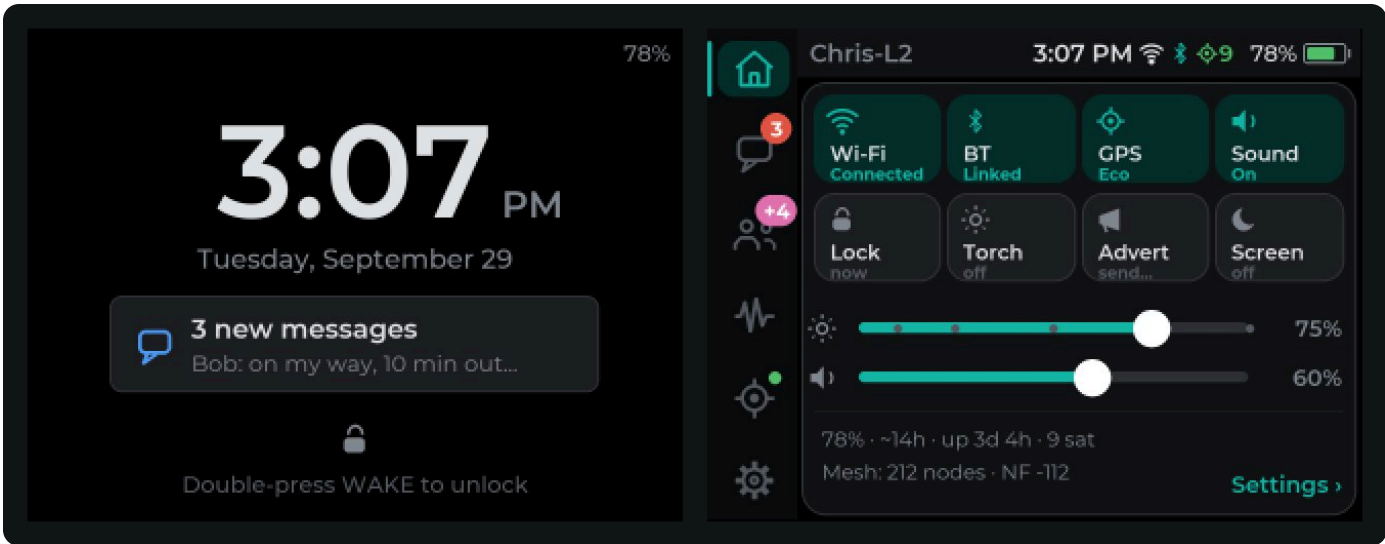
Work through these once, in this order:

- [] Settings → Radio: choose the region for your band, and a TX power no higher than the board's rating for that region.
- [] Settings → Connect → Bluetooth: set a PIN, then pair the MeshCore app if you use one.
- [] Settings → Location → GPS: Eco for days of battery, On for tracking.
- [] Radio → Advert: announce yourself so other nodes can find you.

03 The basics

This chapter covers the screen layout, touch gestures, the buttons, locking, screen timeout and the Quick panel.

[image: The lock screen and the Quick panel]



The lock screen and the Quick panel. Sample data.

Screen layout

Most pages have three areas:

- **Icon rail** down the left edge, with one icon per tab.
- **Status bar** along the top, showing the title, clock and status icons.
- **Content area** below the status bar.

Keyboard pages have no rail, so the status bar spans the full width. The boot splash, lock screen and hibernate screen fill the whole display and have no rail or status bar.

Root pages and sub-pages

- A tab's first page is its root page. The root page shows its title in grey at the left of the status bar.
- A sub-page shows a back chevron "<" before its title. Tap anywhere in that left zone, or just below it, to go back.
- If an editor has unsaved changes, it shows "• unsaved" after its title. When space is short, this becomes just "•".

The rail tabs

From top to bottom, the tabs are Home, Chats, Nodes, Radio, GPS and Settings. The rail shows icons only, and a pill highlights the active tab.

Action	Result
Tap another tab	Opens that tab at its root page.
Tap the current tab while on its root page	Scrolls back to the top.

Action	Result
Tap the current tab while on a sub-page	The first tap returns to the root page and keeps your place in the list. A second tap scrolls to the top.
Hold a tab	Same as a tap.
Hold Chats while you have unread messages	Asks "Mark all read?" with No and Yes . Yes shows "All marked read".

If an editor has unsaved changes, switching tabs first asks "Discard changes?" with **Keep editing** and **Discard**. You cannot swipe sideways to change tabs.

Rail badges and markers

Tab	Marker	Meaning
Chats	Red pill, "1" to "99", then "99+"	Unread messages
Nodes	Magenta pill, "+1" to "+99"	New adverts heard
Radio	Red "PAUSED" pill	A Band scan is using the radio, so the node hears nothing. No other Radio marker shows while it is up.
Radio	Brief amber up-arrow	A packet was sent
Radio	Brief blue dot (right)	A packet was received
Radio	Amber dot (left)	The noise floor is not calibrated yet
GPS	Satellite count (up to 99)	Green: GPS has a fix. Dim: still searching.
GPS	Small moon	GPS Eco mode. The receiver sleeps between fixes.
GPS	No marker	GPS is off
Settings	Amber dot	Update mode is on
Settings	Blue dot	The microSD card is busy (formatting or deleting a map set)

The sent and received flashes only appear while the screen is lit. On keyboard pages the rail is hidden, so the unread count moves to the status bar.

Status bar

The title sits on the left. The clock is in the centre and always stays there. It reads "--:--" until the time is synced and follows your 12-hour or 24-hour setting. Status icons and the battery are on the right.

From right to left, the status icons are:

Icon	Shown when	Meaning
Battery	Always	See "Battery icon" below.
Bluetooth	Bluetooth is on	Blue: the phone app is linked over Bluetooth. Dim: on but not linked.
Wi-Fi bars	Wi-Fi is on and active	The bars show signal strength. White: connected. Blue: the app is linked over Wi-Fi. Amber: update mode. Dim: not connected.

Icon	Shown when	Meaning
Speaker with slash	Sound is muted	—
Padlock	The device is locked	—
Red envelope with count	On keyboard pages only, when you have unread messages	The count stops at "99+".

When space runs short, icons are dropped in this order: Bluetooth, mute, Wi-Fi, then the envelope. The padlock and battery always show. Long titles are shortened with "...".

Battery icon

The battery percentage appears inside the battery outline.

Look	Meaning
Amber fill	On battery, 20 % or less
Red fill	On battery, 10 % or less
Lightning bolt at the tip	Charging
Plug at the tip	Powered by USB, or full
Solid fill reading "100" with a plug	Full
Empty grey outline reading "--"	No battery reading

On USB power, the level is only an estimate. The device has no "charge done" signal.

Touching the status bar

- On a root page, tap the title, clock or status icons to open the Quick panel.
- On a sub-page, tap the clock or anything to its right to open the Quick panel.
- On a sub-page, tap the "◀ Title" area to go back.
- Holding works the same as tapping.

Touch gestures

Gesture	How to do it	What it does
Tap	Touch and release quickly without moving.	Selects or activates. A highlight appears while you press.
Hold	Keep your finger still for about half a second.	Runs the control's hold action while your finger is still down. If a control has no hold action, a hold counts as a tap.
Scroll / drag	Move your finger.	Lists follow your finger. A quick flick keeps a list moving.
Back swipe	Swipe right, mostly sideways. Start well in from the left edge, not at the rail.	Goes back one page. This works even if the swipe starts on a button. It does nothing on root pages, the map or sliders.
Pinch	Use two fingers.	Zooms the GPS map. Everywhere else, a second finger is ignored, so a resting thumb won't cancel a tap.

A short tick sounds when a hold opens a sheet, dialog or page. There is no tick when a hold just acts as a tap. There is also no tick when you hold something inside an open panel or sheet.

Toasts, dialogs and sheets

- **Toasts** are short messages near the bottom of the screen. Each new toast replaces the last one. Some toasts can be tapped, such as "New message from...". Errors have a red bar.
- **Dialogs** have a title, a short message and two or three buttons. The safe choice is always on the left. Tapping outside the dialog, pressing User or locking the device chooses the safe option. Taps are ignored for a moment after a dialog opens.
- **Action sheets** list choices with a Cancel row, which is sometimes labelled differently, for example "Not now". Holding a row counts as a tap. Tapping outside the sheet cancels it.
- Cancelling a dialog does not show a toast.
- While a panel, dialog or sheet is open, it takes every touch, including the rail and status bar.

Physical buttons

User button

User is the configurable button, also labelled **Boot**. You can set separate actions for a click, a double-click and a long press. A long press means holding for 1 s.

Single-click actions start after a short delay. The device waits briefly to see whether you will press again.

Some rules always come first and cannot be changed. They apply in this order:

1. **Locked:** a click or long press lights the lock screen for about 2.5 s and does nothing else. A double or triple press unlocks.
2. **Boot splash showing:** a long press starts the USB rescue command line. You see "Rescue CLI (USB) · restart to leave". The app link and saving stop until you restart. The splash only lasts about 3 s, so avoid holding User during startup.
3. **Screen dark:** any press only wakes the screen.
4. **Flashlight on:** any press only turns the flashlight off.
5. **Triple-click:** toggles mute and shows "Sound on" or "Sound muted". Unmuting plays a confirmation tone.
6. **A panel, sheet or dialog is open:** a press with an action assigned closes it instead of running the action. So if Click opens the Quick panel, a second click closes it. A press set to "None" does not close anything.
7. **During the boot splash:** only the "Back" action runs.

To change the presses, open **Settings** → **Controls** → **Buttons**.

Row	Hint	Default
"Click"	"Press once"	Quick panel
"Double-click"	"Press twice"	New message
"Long press"	"Hold 1 s"	Back

Tap **Save** to keep your changes. You see "Buttons saved". **Cancel** discards them. With the defaults, the Controls card reads "Quick · Msg · Back". If space is short, it uses a shorter form.

Each row can use one of these actions:

Action	What it does
"None"	Nothing.
"Quick panel"	Opens the Quick panel.
"Back"	Closes an open sheet, or goes back one page. On a tab's root page, it turns the screen off.
"New message"	Opens Chats and the recipient picker. It does nothing if the picker is already open.
"Send advert"	Opens the Zero-hop / Flood advert sheet. Nothing is sent until you tap Send .
"Ping last node"	Asks "Ping \?" about the node you last heard from or messaged. Tap Ping to send. The node replies with its round-trip time and signal. Other messages: "No node to ping yet", "Ping in progress...", "Node gone".
"Scan repeaters"	Opens Nodes → Repeater scan and starts a scan if one is allowed. If the Repeater scan page is already open, it just starts a scan. During the cooldown, it shows "Scan again in Ns" and sends nothing.
"Toggle GPS"	Switches GPS from On to Eco, or from Eco or Off to On. Without a GPS receiver, it shows "No GPS receiver".
"Flashlight"	Turns the whole screen white at full brightness. See "Flashlight" below.

If an editor has unsaved changes, "Back", "New message" and "Scan repeaters" first ask "Discard changes?".

The Buttons page reminds you: "Triple-click always mutes, double-click unlocks, and a dark screen only wakes."

Flashlight

- A tap or a User press turns the flashlight off and restores your brightness.
- A WAKE click turns the flashlight off and also turns off the screen.
- Locking, changing pages or a power notice also turns it off.
- It stays on past the screen timeout for up to 10 minutes. After that, the screen times out normally.
- New messages still play a tone, add a badge and show a toast.

WAKE button

WAKE is the side button. You cannot change what it does.

Press	Screen lit, unlocked	Screen dark	Locked, screen lit
Click	Turns the screen off. Also locks if Auto lock is on.	Wakes the screen. If locked, it lights the lock screen for about 2.5 s.	Turns the screen off.
Hold 1 s	Locks now with a tone and "Locked - 2x WAKE". The screen goes dark shortly after.	Wakes the screen. If locked, it lights the lock screen.	Nothing.
Double or triple press	Nothing.	Wakes the screen. If locked, it unlocks .	Unlocks .

Turning the screen off with WAKE stops new messages from lighting it until you wake or unlock it. Messages are still saved, play a tone and add a badge.

WAKE cannot power the device on from hibernate. Only User can.

RST button

RST restarts the device. Afterwards, **About** → "**Last restart**" reads "Reset pin".

To enter the flasher for a firmware update over USB:

1. Hold **User**.
2. Tap **RST**.
3. Release **User**.

Hibernate and powering back on

Open **Settings** → **System** → **Hibernate** to turn the device off until you press User.

1. Hold "**Hold to hibernate**" for about 1.5 s.
2. In the "Hibernate?" dialog, tap **Yes**.
3. Release the screen and all buttons. The device then shows "hibernating..." and "Press User to power back on".

While hibernating, the device receives and relays nothing. Chats and settings, including the GPS mode, are kept. If chat history is off, chats are cleared. Press **User** to power back on.

Lock screen and unlocking

Locking keeps touches and button presses from doing anything by accident.

What the lock screen shows

- A large clock and the date. If the time is not synced, it shows "--:--" and "time not synced".
- The battery percentage at the top right. It turns amber at 20 % or less when not on USB.
- A card when you have unread messages, for example "1 new message" or "3 new messages". It previews the first unread message.
- A padlock with a hint. It reads "Double-press WAKE to unlock", or "Hold the padlock, or double-press WAKE" when Touch unlock is on.

To lock

- Hold **WAKE** for 1 s. The screen goes dark shortly after.
- Tap the "**Lock**" tile in the Quick panel. The lock screen stays lit until the normal timeout.
- Turn on **Auto lock**. Then every screen-off also locks the device, silently. This includes the timeout, a WAKE click, "Back" on a root page and the Quick panel "**Screen**" tile.

Locking plays a tone and shows "Locked - 2x WAKE". Locking also closes any open panel, sheet or dialog. They do not come back when you unlock.

To unlock

- Double-press **WAKE** or **User**. This also works from a dark screen.
- Or, if **Touch unlock** is on, hold the padlock for 1.5 s. A ring fills around it. Sliding your finger cancels.

Unlocking shows "Unlocked" with no tone. Locking or unlocking also lights a dark screen to confirm.

While locked

- Touch is ignored, unless Touch unlock is on.
- Button clicks and holds only light the lock screen.
- A lit lock screen always times out, even if "Screen timeout" is set to "Never".

Wake & lock settings

Open **Settings** → **Controls** → **Lock**. The page that opens is titled "Wake & lock".

Setting	Hint	Default
"Tap to wake"	"Touch the dark screen to light it"	On
"Auto lock"	"Lock when the screen times out"	Off
"Touch unlock"	"Hold the padlock 1.5 s to unlock"	Off

When you save, a toast names the change, for example "Auto lock on". If you changed several settings, it reads "Wake & lock saved". The card summary looks like "Tap on · Auto off".

Screen timeout and wake

Open **Settings** → **Display** → **Display**.

Setting	Options	Default
"Brightness"	10, 25, 50, 75 or 100 %. The slider snaps to the nearest dot when you let go.	75 %
"Screen timeout"	"5s", "10s", "15s", "30s", "1m", "5m", "Never"	15 s
"Stay on while on USB"	On / Off	Off
"Wake screen on message"	On / Off	On

- While you change brightness, the editor shows the new and old values, for example "50% (was 75%)".
- "Stay on while on USB" keeps a lit screen on while USB power or a charger is connected. It does not apply while locked, and it does not wake a dark screen.
- When "Wake screen on message" is off, the Display card shows "no wake".
- Saving shows a toast such as "Screen off after 30s", "Wake on message off" or "Display saved".

Waking the screen

- Press any button.
- Or tap the screen, if **Tap to wake** is on.

The touch that wakes the screen does nothing else. If your finger was on the glass when the screen went dark, lift it before tapping to wake.

When a message arrives

- **You are using the device:** a toast appears if the screen is lit and unlocked. It reads "New message from \"", "\" in \" or "New message in \". Tap it to open the thread. Over the keyboard, the toast cannot be tapped. No toast appears if that thread is already open.
- **The device has been idle, with Wake screen on message on:** the screen wakes and shows a preview of the message.
- **You turned the screen off, or an app is connected:** the screen stays dark. The preview waits for your next wake.
- **The screen stays dark and a draft or open keyboard would be interrupted:** the message is saved and badged instead, and no preview waits for your next wake. If the screen is lit or wakes, the preview still appears on top.
- **No preview appears** when that conversation is already open or update mode is on.

Quick panel

The Quick panel gives you one-tap access to common controls and a summary of your device's status.

Open and close

To open it:

- Tap the status-bar clock or the icons to its right. On a root page you can also tap the title.
- Or press **User** once, if Click is set to "Quick panel" (the default).

The panel does not open while the device is locked.

To close it:

- Press **User**.
- Swipe up on the panel.
- Tap outside the panel.

Locking or an incoming message preview also closes it.

Tiles

The panel has two rows of four tiles.

- A highlighted tile is on.
- An outlined tile is off.
- A dimmed tile is unavailable.

Hold a toggle tile to open its settings. **Back** then returns you to the page you were on.

Tile	Tap	Status shown	Hold
"Wi-Fi"	Turns Wi-Fi on or off. Turning it off asks "Turn Wi-Fi off?" if the app is linked over Wi-Fi or update mode is on. Ending update mode restarts the device.	Connected, App linked, Connecting, No network, Not in range, Low mem, Updating, Off or N/A	Wi-Fi settings
"BT"	Turns Bluetooth on or off. Turning it off asks "Turn Bluetooth off?" if the app is linked.	Linked, On or Off	Bluetooth settings
"GPS"	Cycles On, Eco, Off. Without a GPS receiver, it shows "No GPS receiver".	On, Eco, Off or None	GPS settings
"Sound"	Mutes or unmutes, with no tone.	On or Muted	Sound settings
"Lock"	Locks now.	now	Same as tap
"Torch"	Turns on the flashlight in place of the panel.	off	Same as tap
"Advert"	Opens the Zero-hop / Flood advert sheet. Nothing is sent until you tap Send .	send...	Same as tap
"Screen" (moon)	Turns the screen off. Also locks if Auto lock is on.	off	Same as tap

The toggle tiles show a toast but play no tone. After you answer a "Turn ... off?" dialog, the panel reopens.

Sliders

Slider	Behaviour
Brightness	Changes as you drag and saves when you let go, showing "Brightness N%". It snaps to 10, 25, 50, 75 or 100 %.

Slider	Behaviour
Volume	Changes and saves when you let go, showing "Volume N%". It plays your message tone once, or Chirp if the message tone is set to none. While muted, it reads "Off" and is greyed out. Tapping it then shows "Muted: turn Sound on first".

Tap a slider or drag it sideways. Dragging up or down closes the panel.

Status lines

Line	Example	Notes
1	"78% · \~14h · up 3d 4h · 9 sat"	Shows battery, estimated time left, time since restart and GPS satellites. On external power, the battery part reads "N% · USB". The GPS part may read "GPS eco", "GPS off" or "no GPS".
2	"Mesh: 212 nodes · NF -112"	Shows stored contacts (not nodes heard right now) and the noise floor. "NF --" means the noise floor is not calibrated yet.

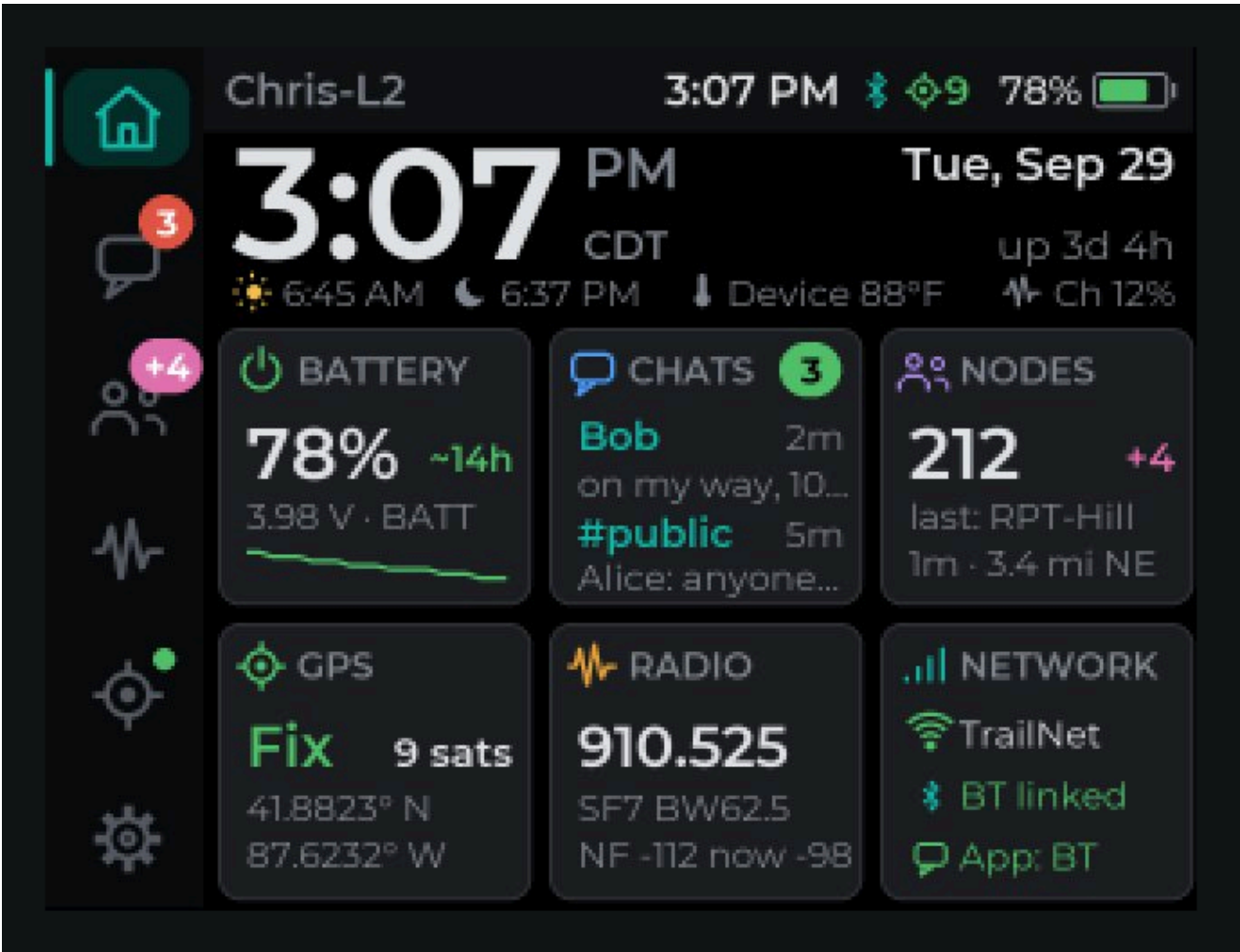
Settings link

Tap "**Settings** ," at the bottom right to close the panel and open the Settings tab. If an editor has unsaved changes, it asks first.

04 Home

Home gives you a one-screen summary of the device: time, conditions, battery, messages, nodes, GPS, radio and connections. Tap a card to open the full screen for that item.

[image: Home]



Home. Sample data.

Opening Home

- Home is the first icon on the left rail, the house.
- The rail order is Home, Chats, Nodes, Radio, GPS, Settings.
- The title at the top shows your node's name. If no name is set, it reads "Home".
- Screens you open from Home, such as the Clock editor or Settings › Connect, keep the Home icon lit on the rail.

Layout

Area	Contents
Header	Large clock, date and uptime

Area	Contents
Strip	Sunrise and sunset, temperature, channel busy
Top row of cards	"BATTERY", "CHATS", "NODES"
Bottom row of cards	"GPS", "RADIO", "NETWORK"

Each card has an icon at its top left and a title in capitals. A card lights up while you press it.

Touch actions

Action	How	Result
Tap	Press and release within about 0.6 s	Opens the card's screen
Long-press a card	Hold about 0.5 s	Same as a tap, except on "RADIO" (opens "Send advert")
Long-press the header or strip	Hold about 0.5 s	Opens the "Clock" editor
Drag or swipe	Start moving on a card	Cancels the press; nothing opens

A long-press acts while your finger is still down. You do not need to lift it.

Header: clock, date and uptime

Time. The large digits show the time, for example "3:07" (12h) or "15:07" (24h).

- In 12h style, "AM" or "PM" appears beside the digits.
- The time-zone abbreviation appears below, for example "EDT", "AKDT" or "UTC+5:30" for a custom offset.
- In "Analog" style, a small clock face appears on the left. The 12h digital time, AM/PM and zone appear beside it.

Clock not synced. The time shows "--:--" in grey, and the date is replaced by "time not synced" in the warning colour.

- The zone abbreviation still shows, but AM/PM does not.
- "Analog" style shows the digital "--:--" layout until the clock is synced.

Date. Top right, for example "Tue, Sep 29".

Uptime. Bottom right, for example "up 12m", "up 4h 12m" or "up 3d 4h".

- With a long custom UTC offset, space is tighter, so the smaller unit is dropped ("up 3d").
- The uptime counter restarts after about 49 days.

The clock updates each minute. After a clock sync or a time-zone change, it updates within about 5 seconds.

Clock editor

Long-press anywhere on the header or the strip to open the "Clock" editor.

Setting	Options	Default
Style	"12h", "24h", "Analog"	"12h"
Time zone	Eastern, Central, Mountain, Arizona, Pacific, Alaska, Hawaii, Custom	—

Setting	Options	Default
"UTC offset" (Custom only)	-12 to +14 h, in 15-minute steps	Offset of the previously selected zone

What the editor shows:

- The "Time zone" caption shows what the zone means right now, for example "EDT · UTC-4".
- For US zones with daylight saving, a "Daylight saving" row shows this year's dates. For example: "Mar 8 – Nov 1, 2 AM, set by itself", plus "On now" or "Off now".
- If the clock is not synced, that row reads "2nd Sun of Mar – 1st Sun of Nov".
- Arizona and Hawaii show "Not observed: the clock never shifts".
- Custom shows the "UTC offset" stepper, captioned "No daylight saving". The stepper wraps around at each end.
- A "Now" sample card previews your choice. Scroll the editor to reach it.
- If the clock is not synced, the sample shows "--:--" and "Not set: syncs from the app or GPS".

Daylight saving is applied automatically for US zones that observe it.

When you save, a message confirms your choice:

- Named zones, for example "12h · Eastern (EDT)".
- Custom, for example "12h · UTC+5:30".

Header strip

The thin strip under the clock holds up to three items. The strip itself has no tap action. It updates about every 5 seconds.

Sunrise and sunset. A gold sun shows today's sunrise. A moon shows today's sunset. Example: "7:21 AM" and "7:02 PM", in your 12h or 24h style.

- Near the poles it shows "Sun up all day" or "No sunrise" (grey).
- It needs a synced clock and a position.
- The position comes from the live or most recent GPS fix. Without one, it uses the advert position set in preferences.
- With neither, the item is hidden.
- It recalculates when the date changes or you move about 1 km.
- Times are accurate to within about one minute.

Temperature. A thermometer and a value, for example "72°F".

- If an environment sensor is attached, its reading is shown.
- Otherwise the processor's own sensor is used, labelled "Device", for example "Device 98°F". This is the chip temperature, not the air temperature.
- Units are °F, or °C when the Units preference is metric.
- The item is hidden if there is no valid reading.
- Sensors are detected only at startup. If a sensor's first answer has no temperature, it is not asked again until you restart.

Channel busy. A radio icon and a percentage, for example "Ch 12%".

- This is the share of the last 60 seconds in which the channel signal was at least 6 dB above the noise floor.
- It shows "Ch --" in grey for about the first 10 seconds.
- The item is hidden on hardware that cannot keep channel history.

Channel busy	Colour
Up to 30 %	Normal
Above 30 %	Warning
Above 60 %	Red

These colours match the BUSY readout on the Spectrum page.

When space runs out, the strip drops items in this order: the word "Device", then the temperature, then channel busy. Sunrise and sunset is always shown.

BATTERY card

Part	Shows
Large number	Charge, for example "78%". On battery, warning colour at 20 % or below
Top right, on battery	Estimated time left, for example "~14h" (green)
Top right, on USB	"chg" while charging, "full" when full, or blank
Line 1	Voltage and source, for example "3.98 V · BATT" or "4.01 V · USB"
Line 2	Trend graph of recent charge

- The time estimate appears only after charge has dropped at least 2 % over at least 10 minutes. A full pack shows none.
- The top-right value is left out if it does not fit beside the large number.
- If the power source is unknown, line 1 shows only the voltage. The card then behaves as if on battery.
- The graph shows "logging..." until it has enough samples. A steady battery draws a flat line.
- Graph colours: green on battery, warning colour at 20 % or below, accent colour on USB.
- The graph restarts when you plug in or unplug.

Charging and full are estimated from the battery voltage. The board cannot report charger status directly. As a result:

- After charging ends, "chg" can show for as long as the charger sits idle.
- "full" can appear while the last few percent are still charging.
- A lasting load change, such as the screen going dark, can show "chg" for a few minutes.
- The percentage reads high while on USB.

Tap: opens the Battery page.

CHATS card

- **Unread count:** a green pill beside the title shows total unread messages, up to "99+".
- **Conversations:** up to the two newest conversations.
- Each conversation shows its name, its age at the right ("45s", "12m", "3h", "2d") and a one-line preview.
- The name is in the accent colour if that conversation has unread messages.
- Your own messages start with "You: ". Channel messages show "Sender: text".
- With no conversations, the card reads "No messages".

Tap: opens the Chats tab.

NODES card

Part	Shows
Large number	Total number of contacts
Top right	" +N" in magenta: new adverts heard since you last opened Nodes, up to "+99"
Line 1	"last: \" , the most recently heard advert
Line 2	How long ago, plus distance and direction when known, for example "1m · 3.4 mi NE"

- Opening the Nodes tab clears "+N". It also stays clear while the Nodes tab is open.
- Distance and direction need your own GPS position, live or last fix. The advert position in preferences is not used.
- The node must be in your contacts with a known position.
- Without positions, line 2 reads for example "5m ago".
- Distance uses mi or km, following the Units preference. Short distances use ft or m.
- Direction uses 8 compass points.
- Before any advert is heard, the card reads "no adverts yet" (grey).

Tap: opens the Nodes tab.

GPS card

State	Large text	Line 1	Line 2
No GPS hardware	"None" (grey)	"no receiver"	—
Fix	"Fix" (green), with "9 sats" at top right	Latitude, for example "41.8823° N"	Longitude, for example "87.6232° W"
On, no fix yet	"No fix" (warning)	Satellites in view, for example "12 in view"	"fix 5m ago", or "searching..."
Receiver silent	"No fix" (warning)	"No GPS data" (warning)	"fix 5m ago", or "check receiver" if there was no earlier fix
Eco, asleep	"Eco"	"next 3:15 PM", "next in Nm" or "sleeping"	"fix Nm ago" or "no fix yet"
Off	"Off" (grey)	"receiver off"	Last fix age, if any

- In Eco mode while awake, line 1 reads for example "eco: 12 in view".
- If the receiver does not report satellites in view, line 1 shows the used count, for example "0 sats".
- Satellite counts go up to 99. "N sats" is left out if it does not fit.
- "No GPS data" means the receiver is powered but has sent nothing for over 10 seconds. It overrides "Fix".
- Possible causes of "No GPS data": wiring, a dead module or a wrong baud rate.
- In Eco, "next 3:15 PM" needs a synced clock. Otherwise it shows "next in Nm".
- GPS modes are off, on and eco.

Tap: opens the GPS tab.

RADIO card

Part	Shows
Large number	Frequency in MHz, for example "910.525"
Line 1	Spreading factor and bandwidth, for example "SF7 BW62.5" or "BW250"
Line 2	Noise floor and current signal in dBm, for example "NF -112 now -98"

- If line 2 does not fit, it shortens to for example "NF -120 / -120".
- "--" means no reading yet.

Tap: opens the Radio tab.

Long-press: opens the "Send advert" sheet on top of Home.

1. Long-press the "RADIO" card.
2. Choose "Zero-hop" ("Neighbors in direct range only") or "Flood" ("Relayed across the mesh (scoped flood)"). "Zero-hop" is selected each time the sheet opens.
3. Press "Send".

The result appears as "Advert sent", "Flood advert sent" or "Advert failed". A successful send also plays a tone.

To close the sheet without sending, press "Cancel", tap outside the sheet or press the User button.

NETWORK card

The card has three lines: Wi-Fi, Bluetooth and App. It always matches the Settings › Connect tile and the quick panel.

Wi-Fi line. When connected, the lit arcs show signal bars.

Text	Meaning	Colour
Network name	Connected; long names end in "..."	Normal
"Connected"	Connected to a hidden network	Normal
"Joining..."	Connecting; only the icon's centre dot is lit	Accent
"Not found"	No saved network worked; tries again later	Warning
"Low mem"	Low memory	Warning
"Updating"	Firmware update mode	Warning
"No network"	Wi-Fi on, no network saved	Grey
"Wi-Fi off"	Wi-Fi turned off	Grey
"No Wi-Fi"	No Wi-Fi hardware	Grey

The Wi-Fi icon is green when connected. It turns accent-coloured when the companion app uses Wi-Fi. In other states it matches the line's colour.

Bluetooth line.

Text	Meaning	Colour
"PIN 123456"	Advertising; this is the pairing PIN	Normal

Text	Meaning	Colour
"Advertising"	Advertising, no PIN	Secondary
"BT linked"	An app is linked	Green text, accent icon
"BT off"	Bluetooth off, or firmware update mode	Grey

The pairing PIN can change each time the device starts. Read the current PIN [here](#).

App line. Shows which link the companion app is using.

Text	Meaning	Colour
"App: Wi-Fi"	A trusted ("Always allow") device	Green
"App: Wi-Fi"	A device allowed once	Accent
"App: BT"	Connected over Bluetooth	Green
"App waiting"	A Wi-Fi app is waiting for you to allow it	Warning
"No app"	No app connected	Grey

If the app is connected over both Wi-Fi and Bluetooth, the card shows Wi-Fi.

Tap: opens Settings › Connect. Press Back to return to Home. The Wi-Fi and Bluetooth rows there open their editors. The Bluetooth PIN setting is in the Bluetooth editor.

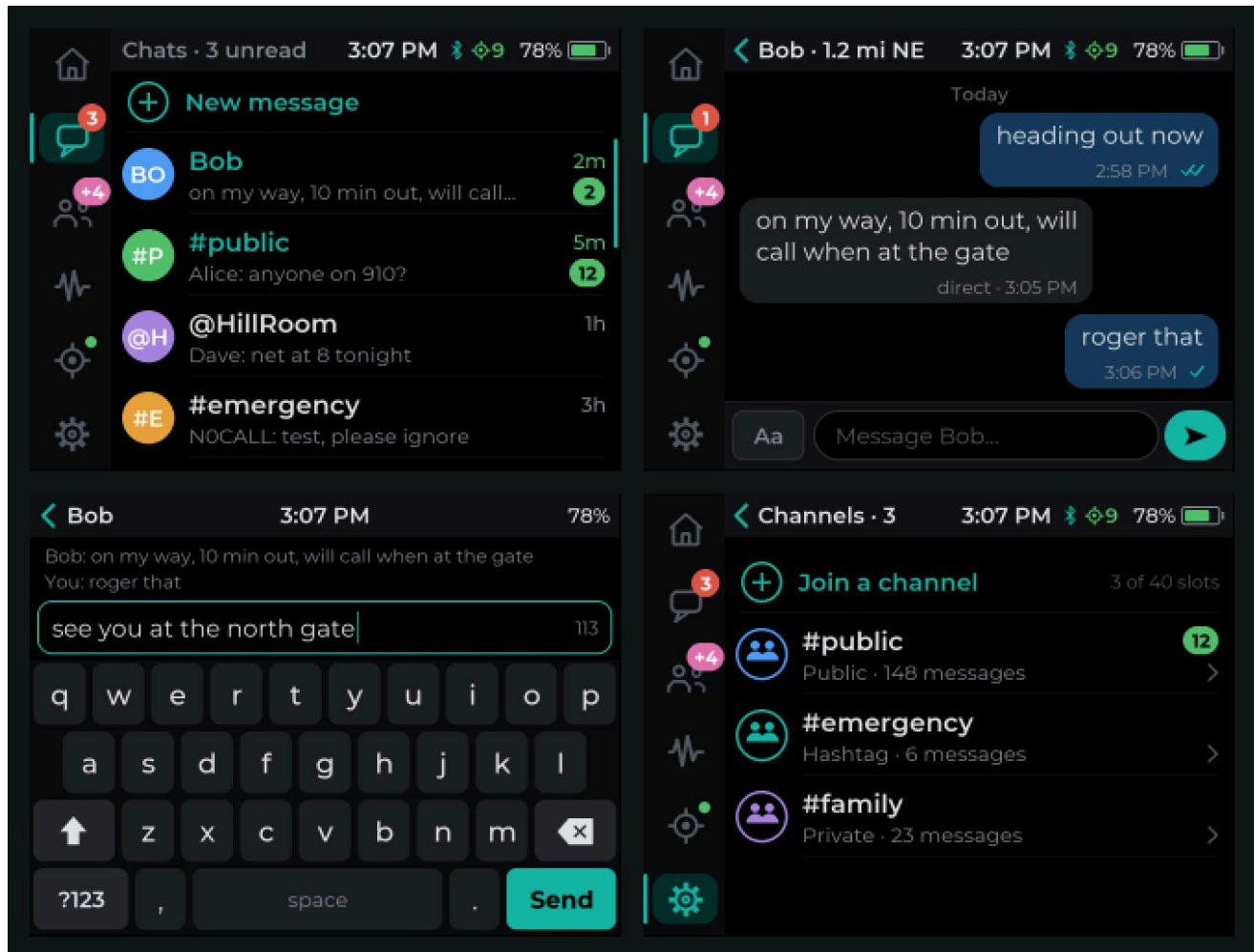
When Home updates

- Home updates only while it is shown and the screen is on.
- When you return to Home or the screen wakes, everything refreshes at once.
- Battery and Radio update about every 2 seconds. Network updates every second. GPS, the strip and chat ages update about every 5 seconds.
- The Chats card also updates as soon as a message arrives.

05 Chats and messaging

You send and read text messages from the Chats tab: direct messages to contacts, posts to channels, and posts to room servers you are logged in to.

[image: Chats, a conversation, the keyboard and Channels]



Chats, a conversation, the keyboard and Channels. Sample data.

Ways to start a conversation

Where	What to do
Chats tab	Tap "+ New message ", then pick a recipient.
User button	Double-press. By default this runs "New message" and opens the recipient picker.
Nodes tab	Long-press a node, then tap " Message ". Nodes you have only heard cannot be messaged.
Node detail page	Tap " Chat ". If it is greyed out, a toast says "Can't message this node" or "Not a saved contact".
Channel page	Settings → Radio → Channels → pick a channel → " Open chat ".
New-message toast	Tap a toast such as "New message from Bob".

If you have unsaved changes in an editor elsewhere, the device may first ask whether to discard them.

Repeaters are never chat destinations. Their Node detail button reads "Admin" instead of "Chat".

Default User button actions

Press	Action
Single press	"Quick panel"
Double press	"New message"
Hold	"Back" (at a tab root, this turns the screen off)
Triple press	Toggle mute

You can reassign these in Settings → Controls → Buttons. The card summary reads "Quick · Msg · Back". When a sheet, dialog or the quick panel is open, any mapped press just closes it.

Chat list

The title reads "Chats", or "Chats · N unread" when you have unread messages. The top row is **" + New message "**. Conversations follow it, newest first.

Row element	Meaning
Coloured avatar	The contact's or channel's initials.
Name	Shown in accent colour while the conversation has unread messages.
Age	Time since the newest message: "12s", "5m", "3h", "2d". Green while unread.
Preview	The last message on one line. Your own reads "You: ..."; channel posts start with the sender's name.
Green pill	Unread count; "99+" above 99.

A name the device cannot resolve shows as "Unknown". Channel names always start with "#", for example "#Public". With no conversations, the list shows "No messages yet." / "Tap + to send one."

The list does not re-sort while your finger is on it. It updates when you lift.

Action	Result
Tap a row	Opens the thread.
Long-press a row	Opens a sheet: "Open", "Mark conversation read", "Mark ALL read...".
Tap the Chats tab at the list	Scrolls to the top.
Tap the Chats tab inside a thread	Returns to the chat list.

"Mark ALL read..." asks "Mark all read?" with [No] [Yes]. No is the default. Yes shows "All marked read".

Unread indicators

- **Rail badge:** a red pill above the Chats icon shows the total unread count ("99+" above 99).
- **Status bar envelope:** shows the count while the keyboard hides the rail.
- **Rail long-press:** long-press the Chats tab to get "Mark all read?". With nothing unread, it just opens the tab.

Messages become read when:

- You open their thread. The whole conversation is marked read.
- You view them in the Reader.
- The connected phone app reports it has read everything.

Messages that arrive while a sheet, dialog or the flashlight covers the thread stay unread until you close it. Your own sent messages never count as unread.

Choosing a recipient

The **"New message"** picker lists everything you can message.

1. Tap **" + New message "**, or double-press the User button.
2. Optionally tap **"Search..."** and type part of a name. Matching ignores capitals. Tap "x" to clear the search.
3. Tap a row. The thread opens with the keyboard already up.

Section	Contents
"CHANNELS · N"	Your joined channels, in slot order.
"CONTACTS · N"	Chat contacts, plus room servers you are logged in to (shown as "@RoomName"). Sorted by name, so rooms appear first.

Contact rows show the route, when known, plus last heard and distance, for example "direct · 14m · 0.8 mi". Here "direct" means the stored route to that contact has no hops. Repeaters are never listed.

Empty states: "No contacts yet" / "Contacts appear as their adverts are heard." and "No matches".

Reading a thread

The title shows the name and message count, for example "Bob · 14 msgs" or "#Public · 1 msg". Oldest messages are at the top. The thread opens at the newest message and stays there as new ones arrive.

Element	Description
Incoming bubble	Left side, card colour. Channel posts show the sender's name above the text.
Your bubble	Right side, blue.
Date divider	"Today", "Yesterday" or a date such as "Tue Sep 29". Appears only when the clock is set.
"..." at the end	The message is too long for the bubble. Tap it to read it all.

Times are the moment your device received the message, not when it was sent. Before the clock is set, times and date dividers are hidden.

Meta line and delivery marks

Each bubble has a meta line at the bottom right.

Bubble	Meta line	Example
Incoming	Hops · time	"direct · 3:05 PM", "1 hop · 3:05 PM", "2 hops · 3:05 PM"
Your own	Time + one tick	"3:05 PM ✓", or "sent ✓" without a clock

In bubbles, "direct" means the message arrived direct-routed. Without a clock, incoming bubbles show only the hop text. The single tick means "sent from this device". There is no second "delivered" tick and no read receipt. The device cannot confirm that a message arrived.

Thread gestures

Gesture	Result
Tap a bubble	Opens the Reader.
Long-press a bubble	Opens a sheet with "Reply with preset..." and "Details".
Swipe right, or tap "◀"	Goes back to the chat list.

The sheet's title is "Your message", the sender's name, or the conversation name.

"Details" opens "Message details" with three lines:

- Who: "To Bob", "From (2) Bob", "From (D) Bob" (direct) or "From Alice in #Public".
- When: "3:05 PM, Tue Sep 29 · 12m ago", or "12m ago" without a clock.
- Route: "Sent from this device" or "Received · 2 hops".

Tap [Open] to go to the Reader, or [Close].

An empty thread shows "No messages yet" with "Say hello to Bob" or "Say hello in #Public".

The Reader

Tap any bubble to see the full message.

- The title shows hops, sender and position, for example "(2) Bob · 3/14". "(D)" means direct. Your own messages show "To Bob · 3/14".
- The second line reads like "Tue Sep 29 · 12m ago · 2 hops", or "sent here" for your own messages.
- Drag up and down to scroll long text.

Button	Action
"◀ Prev"	Older message. Disabled at the oldest.
"Reply"	Pick a quick reply, then confirm.
"Next ▶"	Newer message. Disabled at the newest.

Each message you view is marked read. If storage drops the message while you are reading it, you see "Message no longer stored".

Composing a message

The composer at the bottom of the thread has three parts: **[Aa]**, the message field, and the Send arrow.

1. Tap the field. The full-screen keyboard opens.
2. Type your message.
3. Tap **"Send"** on the keyboard, or close the keyboard and tap the Send arrow.

Field shows	Meaning
"Message Bob..."	Empty draft.
"Recipient gone"	You can no longer send to this conversation.
Leading "..."	Your draft is long; only its end is visible.
A number	Bytes left, shown at 20 or fewer. Warning colour, then red when you are over the limit.

Message limits

Conversation	Limit
Direct message	160 bytes
Channel	160 bytes minus your node name length minus 2

Channel posts are sent as "YourName: text", so a longer name leaves less room. Emoji and accented letters use more than one byte each.

The Send arrow is active only when the draft has text and the recipient still exists.

After you send

On success, the "Sent / confirm" tone plays, the draft clears, and the thread scrolls to your message.

Toast	Meaning
"Sending..."	Direct message queued for sending. Delivery is not confirmed.
"Message sent"	Channel post sent.
"Message too long" (red)	Over the limit, for example after your node name got longer.
"Recipient gone" (red)	The contact was removed, a room login lapsed, or the channel was left.
"Channel changed" (red)	The channel slot now holds a different channel, so nothing is sent. Usually seen from "Reply with preset..." or the Reader's "Reply". In an open thread the device soon shows "Recipient gone" instead.
"Send failed" (red)	The send did not go out.

No tone plays when a send fails. While a thread is open, the device rechecks the recipient every few seconds. If it disappears, you see "Recipient gone" and Send greys out.

Drafts

Each conversation keeps its own unsent draft, for up to 8 conversations. Leaving the keyboard with Back keeps the draft. An incoming-message preview also keeps it. Drafts are lost when the device restarts.

You cannot edit or delete a single message after sending it.

Keyboard

The keyboard fills the screen and hides the rail. The status bar shows "◀ Bob". The last two messages appear above the field for context.

The cursor always sits at the end of your text. You cannot move it.

Row	Keys
1	q w e r t y u i o p
2	a s d f g h j k l
3	Shift, z x c v b n m, Backspace
4	"?123", emoji (smiley), ",", "space", ".", "Send"

Typing

- A key is entered when you lift your finger. You can slide to correct a near miss first.
- A slide that ends off the keys enters nothing.
- **"Send"** fires only if your touch also started on Send.

Shift

- Tap once for one capital letter.
- Double-tap quickly for caps lock. Tap again to turn it off.
- Shift and caps lock reset each time the keyboard opens.

Symbol layers

Key	Shows
"?123"	Numbers and common punctuation: - / : ; () \$ & @ " . , ? ! ' "
"#+="	More symbols: [] { } # % ^ * + = _ \ ~ < > € £ ° •
"123"	Back to the numbers layer.
"ABC"	Back to letters.

Backspace deletes as soon as you touch it. Hold it to repeat. A whole emoji is removed in one press.

Closing the keyboard

- **"Send"** closes the keyboard and sends.
- To keep the draft without sending, hold the User button (Back) or tap "<". A single press opens the quick panel instead.
- The back swipe does not work on the keyboard screen.

Emoji

Tap the smiley key to open the emoji panel.

Tab	Contents
Recent	Up to 24 emoji you used. Cleared on restart.
Smileys, People, Nature, Food, Travel, Objects, Symbols	The full set of 229 emoji.

- The panel opens on the tab you last used, or Smileys the first time.
- Drag or flick the grid to scroll. A touch stops scrolling.
- The bottom row has "ABC", space, Backspace and the action key.

- An emoji that would pass the byte limit is not inserted.
- Skin tones are not supported; hands show in the default yellow.
- An unsupported emoji shows as a small rounded box.

Quick replies

Quick replies are 10 fixed presets. You cannot edit them: "Roger", "Yes", "No", "Standby", "On my way", "Arrived", "Call me", "Need assistance", "Cannot talk now" and "Signal check?".

They appear as a two-column grid in a sheet. Tap one to pick it. Tap outside the sheet or press the User button to close without choosing.

Opened from	What happens
[Aa] in the composer	The preset is added to your draft. Nothing is sent.
"Reply with preset..." on a bubble	Confirm "Send reply?" with [Send] or [Cancel].
"Reply" in the Reader	Confirm "Send reply?" with [Send] or [Cancel].

If a draft exists, [Aa] adds the preset after a space. In a channel, this can push the draft over the limit. The counter then turns red and negative, and Send shows "Message too long".

After a confirmed reply, you see "Sending..." for a direct message or "Reply sent" for a channel.

New-message alerts

When a message arrives, the device either shows a short toast or a full-screen "New message" preview.

- **Toast:** shown if you touched the screen or pressed a button in the last 30 seconds (with the screen on and the device unlocked), or the flashlight is on. Tap it to open the thread. Over the keyboard, it cannot be tapped.
- **Preview:** shown otherwise. It shows the sender, route and message text.
- **Nothing:** if that conversation's thread is already open (unless the flashlight is on, in which case the toast still shows), or during a firmware update.

Preview button	Action
"Dismiss"	Leaves everything unread and returns to where you were.
"Next ›"	Marks this message read and shows the next. Reads "Done" on the last one.
"Open"	Marks it read and opens the thread.
"All read"	Asks to mark every unread message read.

The preview closes by itself once nothing is unread. If "Wake screen on message" is off (Settings → Display), the screen does not wake and no preview appears.

Tone (Settings → Sound)	Default
"Direct message"	Chirp
"Channel"	Kerplop
"Sent / confirm"	Tick

Tone choices are None, Chirp, Kerplow, Tick, Advert, Ding, Double, Rising, Falling, Bell and Trill. Incoming message tones do not play while a phone app is connected.

Message display options

These are in Settings → Display, section "MESSAGES & UNITS".

Option	Values	Default	Effect
"Text size"	"Normal", "Small"	"Normal"	Text size in bubbles, the Reader and the preview.
"Compact messages"	On, Off	Off	One line per message: "3:05 PM Bob: text". Without a clock, the age replaces the time.
"Wake screen on message"	On, Off	On	Whether a new message wakes the screen.

In compact mode, tap a line to read it in full. Long-press still opens the reply and details sheet.

Direct messages, channels and rooms

Type	Who receives it	Limit	Send toast
Direct message	One contact	160 bytes	"Sending..."
Channel	Everyone with the channel's key	160 bytes minus name length minus 2	"Message sent"
Room	A room server you are logged in to	160 bytes	"Sending..."

- Rooms appear as "@Name". You must log in first, using "Log in..." on the contact sheet. Without a login, sending gives "Recipient gone".
- Room posts always appear to come from the room. The original author is not shown.
- Messages you send from the phone app do not appear in the device's chat list.

Channels

You manage channels in Settings → Radio → Channels. The Radio settings card shows "None joined" or "N joined". The device holds up to 40 channels.

The list title reads "Channels · N". The top row is "+ Join a channel" with "N of 40 slots". Each channel row shows a coloured group icon, the name, an unread pill, and a line such as "Hashtag · 12 messages".

Type	Icon colour	Key	Who can read it
Public	Blue	The well-known key	"Everyone on the mesh can read it."
Hashtag	Accent	Made from the name, including "#"	"Anyone who knows the name can read it."
Private	Violet	A random key shared by hand	"Only people with the key can read it."

With no channels, the list shows "No channels." / "Tap + to join one."

Joining a channel

1. Tap "+ Join a channel".

2. Choose a mode: **"Hashtag"**, **"Private"** or **"Public"**.
3. Fill in the fields for that mode (below).
4. Tap **[Join]**.

On success you see "Joined #name". No tone plays.

Hashtag

- Type the name after "#", with no spaces. The same name gives the same channel.
- Capitals count: "#Test" and "#test" are different channels. A name with capitals shows "Has capitals: others must type it the same".
- The resulting key is shown under "Key, from the name".

Private

- **"Name"**: what this device calls the channel, up to 31 bytes. The key decides who is in it.
- **"Key"**: 32 hex digits (spaces are fine) or base64, from whoever set up the channel.
- Tap **"New key"** to create a random key for a new channel. Share the key only with members.

Public rejoins the default channel every MeshCore node starts with.

If you tap [Cancel] or Back after typing, the device asks "Discard changes?".

Join error	Fix
"Give the channel a name"	Enter a name.
"Type the key, or tap New key"	Enter a key or create one.
"Key: 32 hex digits or base64"	Check the key format.
"An all-zero key is not a channel"	Use a real key.
"Already joined as #x"	You already have this channel.
"All 40 channel slots are in use"	Leave a channel first.
"Join failed"	Try again.

Channel page

Tap a channel in the list to open its page. It shows the type, the slot number and how many messages are stored.

Key card label	Type
"Key (well known)"	Public
"Key (from the name)"	Hashtag
"Key · tap to show"	Private. Hidden as dots; tap to show or hide. Once shown, it reads "Key · share only with members".

Tap **"Open chat"** to open the channel's thread.

Leaving a channel

1. Open the channel page.
2. Hold **"Hold to leave channel"** for about 1.5 seconds.
3. Confirm "Leave #name?" with **[Leave]**, or tap [No].

You see "Left #name". Stored messages stay in Chats, but sending to them gives "Recipient gone".

Private channels: the key is not kept after you leave. To rejoin, you need the key again.

You can rejoin Public from Channels and a Hashtag channel by its name at any time.

If the phone app changed the slot meanwhile, you see "Channel changed: not left". The page may also show "No longer in this slot (changed in the app)."

Message storage

The device stores up to 200 messages in total, across all conversations. When full, the oldest are removed first.

"Keep across restarts" (Settings → System → Restart, section "CHAT HISTORY") is on by default. It saves messages and their read state to the device's internal storage. A crash or power cut can lose about the last 30 seconds.

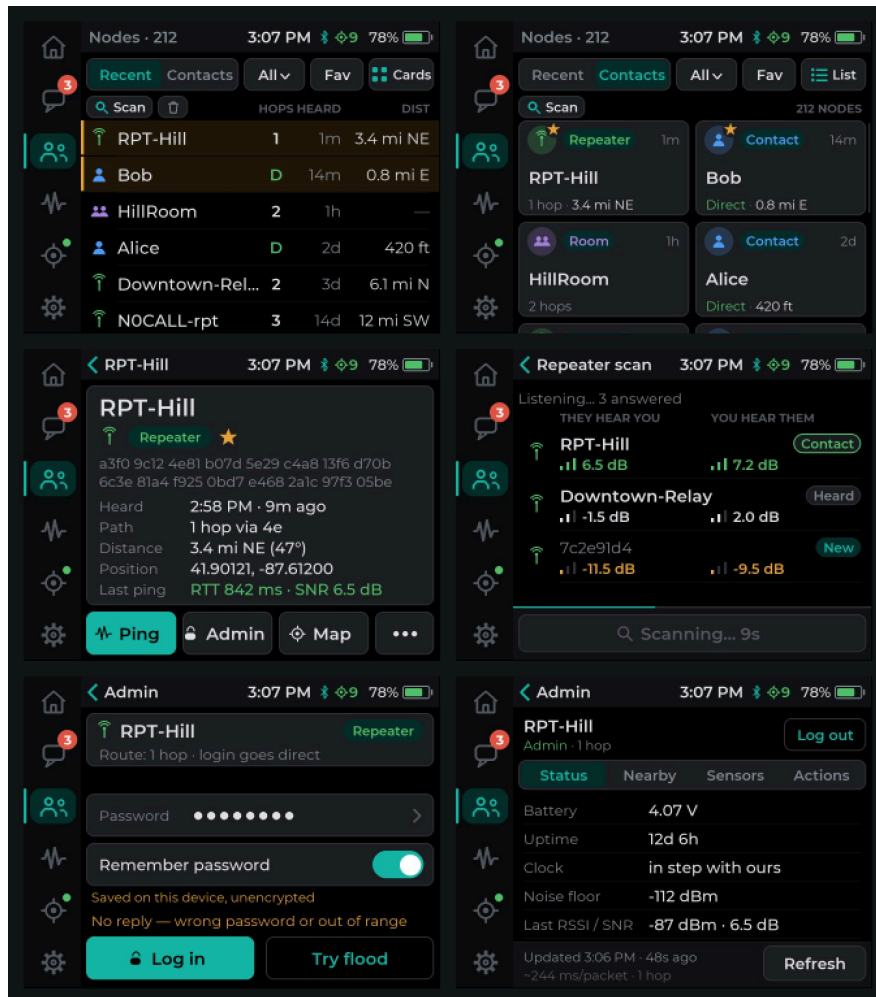
Messages are stored unencrypted. To opt out, turn "Keep across restarts" off and confirm "Stop keeping chats?". The saved copy is erased at once.

To erase all messages, hold **"Hold to clear chat history"** and confirm "Clear chat history?" with [Clear]. This cannot be undone.

06 Nodes

The Nodes page lists the nodes your device has heard or saved, and lets you ping, message, manage and administer them.

[image: Nodes as a list and as cards, Node detail, Repeater scan, and repeater admin (log in and status)]



Nodes as a list and as cards, Node detail, Repeater scan, and repeater admin (log in and status). Sample data.

Opening Nodes

- **Side rail:** the Nodes tab shows a magenta "+N" badge for adverts heard since you last looked. The badge stops at "+99".
- **Home:** the Nodes tile shows your contact count and any new adverts as "+N" in magenta.
- **Map:** the map's List button opens Nodes as a one-time view of Contacts, filter All, sorted by Distance.

When you open Nodes, the new-advert count clears. Adverts that arrive while the page is open count as seen.

When you are already on Nodes, tap the Nodes tab again to scroll back to the top.

After you use the map's List button, Nodes returns to your own mode, filter and sort the next time you open it normally. If you change any of them during the one-time view, your change stays.

Title and fill banner

The title reads "Nodes · N", where N is the number of saved contacts. Your device holds up to 350 contacts. When the table is full, "full" appears in the title in warning color.

A fill banner appears in the header row when the contact table is nearly full.

Banner	When it shows	Tap result
None	Below 90% full	—
"330/350" (dim)	90% full or more, with at least 5 contacts saved	Toast "Contacts nearly full"
"350/350 · FULL" (warning color); "350/350" in Recent mode	Table full	Toast "Full · new nodes aren't saved", or "Full · oldest non-favorites replaced" if auto-add "overwrite oldest" is on

Toolbar

The toolbar runs across the top of the page.

Control	Shows	What a tap does
Recent / Contacts toggle	"Recent" or "Contacts"	Switches the list and scrolls to the top
Filter chip	"All", "Cont", "Rprr" or "Room"	Opens the "Show" sheet
Sort chip	"Heard", "Name", "Dist" or "Fav"	Opens the "Sort by" sheet
View switch	"Cards" in List view; "List" in Cards view	Switches view and toasts "Cards view" or "List view"

Under the toolbar, the header row holds these controls:

Control	Where	What it does
"Scan" chip (magnifier)	Always	Opens "Repeater scan"
Trash chip	Recent mode only	A tap toasts "Hold to clear recents". A hold offers to clear the Recent list.

Recent vs Contacts

- **Recent** lists recently heard adverts, newest first. It holds up to 16 nodes and can include nodes you have not saved.
- **Contacts** lists your saved contacts.

Nodes you have heard but not saved never show a distance or position. Their DIST shows "—", and a Distance sort puts them last.

In Contacts mode, the HEARD age and the "Last heard" sort use the time stamped by the remote node's own clock. If that clock is wrong, ages can look odd or show "-".

Filter and sort

Tap the filter chip to choose which node types to show.

Option	Shows
"All" (default)	Every node, including sensors and nodes you have not saved

Option	Shows
"Contact"	Saved chat contacts only
"Repeater"	Saved repeaters only
"Room"	Saved room servers only

Unsaved nodes have an unknown type, so only "All" shows them. In Recent mode, any other filter shows only nodes that are also saved.

Tap the sort chip to choose the order.

Option	Order
"Last heard" (default)	Newest first
"Name"	A–Z, ignoring case; ties go to the most recently heard
"Distance"	Nearest first; nodes with no known distance go last
"Favorites first"	Favorites at the top, then by last heard

Mode, filter and sort reset after a reboot to Recent, All and Last heard. Only your List or Cards choice is kept.

List view and Cards view

Tap the view switch to change between List and Cards. The node at the top stays in view when you switch. The view you leave goes back to the top.

List view

Each row shows a type icon, the name, and three columns.

Item	Values
Type icon	Person (link color) = contact; antenna (green) = repeater; group (violet) = room; signal (amber) = sensor; grey person = heard, not saved
Name	Cut with "..." if too long
"HOPS"	"D" (green) = direct; a number = hops; "-" (grey) = no known path (flood)
"HEARD"	Age, such as "45s", "12m", "3h" or "2d"; "-" if never heard or its clock is ahead of yours
"DIST"	Distance and bearing, such as "1.2 mi NE"; "—" with no GPS fix or no shared position

Favorites have an amber tint and an amber bar at the left edge. List rows show no star.

The list updates every 2 seconds, but not while you touch or scroll it. Ages update every 5 seconds.

Cards view

The header shows "N NODES" instead of column titles. Each card shows:

- **Top line:** a colored disc with the type icon, a type pill, and the age. Favorites have an amber star on the disc.
- **Type pill:** "Contact", "Repeater", "Room", "Sensor" or "Not saved".
- **Name:** the full name on one or two lines. Only very long names are cut with "...".
- **Meta line:** "Direct" (green), "1 hop" or "3 hops", or "Flood" (grey), then " · " and the distance with bearing.

Tap, hold, mode, filter and sort work the same as in List view. Signal strength (SNR) per node is not shown in either view.

Empty and error messages

Message	Meaning
"No recent adverts"	Recent list is empty
"No contacts yet"	You have no saved contacts
"No matching nodes" / "Change the filter to see more"	The filter hides every node
"Out of memory" (red)	The node lists could not load

Clearing the Recent list

1. In Recent mode, hold the trash chip.
2. In "Clear recents?", tap **Clear**. **No** is the default.

The toast "Recents cleared" confirms it. Saved contacts are not affected.

Tapping and holding a node

- **Tap** a node to open its Node detail page. If it has gone, the toast reads "Node gone".
- **Hold** a node to open an action sheet titled with its name.

Node	Sheet items
Heard, not saved	"Ping", "Details"
Saved repeater	"Ping", "Details", "Admin...", "Contact..."
Saved contact or room you can message	"Ping", "Message", "Details", "Contact..."
Any other saved node (such as a sensor)	"Ping", "Details", "Contact..."

If the contact was removed meanwhile, the toast reads "Contact gone".

Ping

A ping checks that a node can hear you and measures the round trip. It uses the route the node's advert came by, or the contact's saved route.

Toast	Meaning
"Ping sent..."	Ping sent (silent)
"Ping in progress..."	A ping is already out; only one at a time
"RTT 123ms SNR 7.5dB"	The node answered; one acknowledgement tone plays
"Ping timeout"	No answer (silent)
"Ping failed"	The ping could not be sent

Node detail

The title is the node's name, or "Node" if it has none. The identity card shows:

- the full name;
- a type pill, with an amber star for a favorite;
- the public key in groups of 4 hex digits: the full key for a saved contact, or a short prefix ending in "..." for an unsaved node.

Row	Example values
"Heard"	"3:07 PM · 12m ago" (follows your 12/24 h setting); "never"; "its clock is ahead of ours" (warning color)
"Path"	"direct"; "1 hop via 4e"; "3 hops via 4e › a1 › 07" (listed from your side); "unknown · flood"
"Distance"	"1.2 mi NE (47°)"; "no GPS fix here"; "—" if the node shares no position
"Position"	Coordinates to 5 decimals, or "not shared"
"Last ping"	"waiting for reply..."; "RTT 123 ms · SNR 7.5 dB" (green); "—"

The page refreshes every 5 seconds.

Button	What it does	When greyed out
Ping	Pings the node	—
Chat	Opens the conversation	Node can't be messaged: "Can't message this node" (saved) or "Not a saved contact"
Admin (repeaters, replaces Chat)	Opens remote admin	Not saved: "Not a saved contact"
Map	Opens the GPS map centered on the node	No position: "No position shared"
...	Opens the contact sheet	Not saved: "Not a saved contact"

Remote admin needs the node's full key, so you must save a repeater before you can log in.

Contact sheet

Open the contact sheet from ... in Node detail, or from "Contact..." in the hold sheet. It is titled with the contact's name.

Item	What it does	Toast
"Favorite" / "Unfavorite"	Marks or unmarks a favorite	"Favorite: Bob" / "No longer a favorite: Bob"
"Reset route"	Forgets the saved route; no confirmation	"Route reset · next message floods", or "No route saved (already flooding)"
"Remove contact..."	Opens the Remove page	—
"Log in..." (rooms only)	Opens room login	—

When the table is full and "overwrite oldest" is on, favorites are never overwritten.

Contact changes are saved about 5 seconds after your last edit. A crash in that time loses the change.

Locking the device closes any open sheet or dialog.

Removing a contact

The Remove page shows the contact's type, favorite status, short key and number of stored messages. It explains what happens:

- "Its messages stay in Chats."
- Whether its next advert may add it back, depending on auto-add.
- "A connected phone app may list it until it reloads its contacts."

To remove a contact:

1. Open the contact sheet and tap "Remove contact...".
2. Hold the red **"Hold to remove contact"** button for 1.5 seconds. Sliding your finger cancels the hold.
3. In "Remove Bob?", tap **Remove**. **No** is the default.

The toast "Removed Bob" confirms it. The page and the node's detail page close. Any saved admin password for that node is forgotten.

If the phone app already removed the contact, the page reads "Already removed (by the app?)" and the button is disabled. If the contact disappears during the dialog, the toast reads "Contact gone: nothing removed".

Repeater scan

Repeater scan finds the repeaters in direct radio range and shows how well each direction works.

The scan sends one request that is not relayed. Only repeaters in direct range answer. Repeaters with forwarding off, or that answered 4 requests in the last 2 minutes, stay silent.

To run a scan:

1. Tap the **"Scan"** chip on Nodes. If your User button is set to "Scan repeaters" (Settings → Controls → Buttons), pressing it opens this page and starts a scan.
2. Tap **"Scan for repeaters"**.
3. Wait while the button counts down "Scanning... 12s". Listening lasts 15–45 seconds, depending on radio settings.

One acknowledgement tone plays at the first answer. If the request can't be sent, the toast reads "Scan failed".

Button	Meaning
"Scan for repeaters"	Ready
"Scanning... 12s" (disabled)	Listening for answers
"Scan again in 23s" (disabled)	30-second cooldown
"Scan again"	Ready again

Each row shows the repeater's name, a status chip, and signal for both directions: "THEY HEAR YOU" and "YOU HEAR THEM".

Item	Values
Name	Contact name, else heard name, else 8 hex digits in grey

Item	Values
Status chip	"Contact" (saved); "Heard" (in recent adverts, not saved); "New"
Signal bars	3 (green) at 5 dB or more; 2 from -5 dB; 1 (amber) below -5 dB
SNR	For example "-12.2 dB"

Rows are sorted by the weaker direction, then the stronger. Up to 20 of the strongest repeaters are kept.

If no repeater answers, the page reads "No repeaters answered". They may be out of direct range.

Results stay while the page is open. They are cleared the next time you open the page, unless a scan is still listening.

Adding a repeater from the scan

Tap a saved repeater to open its Node detail. Tap any other repeater for a sheet:

- **"Add as contact"** saves it as a repeater with a direct route.
- **"Add and log in..."** saves it, then opens the admin page.

Its real name and position fill in at its next advert.

Toast	Meaning
"Added Hill Top"	Saved
"Already a contact"	Already saved ("Add as contact" only)
"Contacts full · remove one first"	Table full, even with overwrite-oldest on
"Add failed"	Could not save

Repeater remote admin

Remote admin lets you log in to a saved repeater or room server to read its status and send fixed commands.

Open it from:

- **Admin** in Node detail;
- "Admin..." in the hold sheet;
- "Log in..." in a room's contact sheet.

The title is "Admin" for a repeater, or "Room login" for a room.

Logging in

Your device clock must be set before you can log in. Otherwise the page shows "Clock not set · logins will be ignored", and **Log in** is greyed out.

1. Tap the "Password" row and enter the password (up to 15 characters). Leave it blank for guest or saved access.
2. Turn on **"Remember password"** if you want it saved on this device.
3. Tap **Log in**. The status reads "Logging in... attempt 1 of 2". It retries once automatically.

The header card shows the route: "Route: 2 hops · login goes direct" or "No route · login floods".

The password field shows dots only. Saved passwords are stored unencrypted, as the note under the toggle warns.

Result	What you see
Success	"Logged in as admin", "Logged in as guest", or "Logged in · Read-only" / "Read/write"; one tone
No answer	"No reply — wrong password or out of range"; toast "No reply from repeater"
Phone app logging in	"Busy · already logging in, try again"
Send error	"Send failed"

A repeater never says "wrong password". It simply does not answer. If a direct login fails, tap **Try flood** to reset the route and flood the login.

Logged-in page

The header shows the name, your role ("Admin", "Read/write", "Read-only" or "Guest") and the route. Route words are "direct", "N hops" or "no route · sends flood".

If the remote clock is 2 minutes or more off, "clock 3m behind" or "ahead" replaces the route in warning color.

After 2 unanswered requests in a row, the header reads "No replies · tap here to log in again". The repeater may have rebooted.

Tabs:

- Repeater: "Status", "Nearby", "Sensors", "Actions".
- Room: "Status", "Sensors", "Actions".

Status loads once after login. To load any other tab, tap **Refresh** in the footer. The footer shows when the data was updated and the estimated airtime per request.

Request limits

These limits protect shared airtime.

Limit	Message
One request at a time, including pings	"Busy · one request at a time"
6 sends, one more every 20 s; a flood counts as 2	"Wait Ns · saving airtime"
Cooldown after each exchange: at least 10 s direct, 5 s for actions, 30 s flood, up to 120 s	Refresh shows "Wait 20s"
Clock not set	"Set the clock first (GPS, Wi-Fi or app)"

Reads retry once automatically, then show "No reply".

Status tab

Row	Notes
"Battery"	Volts, or "not measured"
"Uptime"	—
"Clock"	"in step with ours", or "3m behind ours" / "ahead of ours" (warning color)
"Noise floor"	—

Row	Notes
"Last RSSI / SNR"	—
"TX queue"	Warning color at 4 or more
"Airtime"	TX and RX (room: TX only)
"Channel use"	Percent
"Received" / "Sent"	Total, then F (flood) and D (direct)
"Duplicates"	F and D
"Radio errors"	"none", or the error types in warning color
"Bad packets"	Repeater only
"Posted" / "Pushed"	Room only
"Firmware"	Repeater only; tap to load firmware, name and owner

Nearby tab (repeaters only)

Nearby lists the repeaters this repeater hears directly. Each row shows a name, SNR with bars, and how long ago it was heard.

- Tap the order row to switch between "Strongest first" and "Newest first".
- Tap "Load more (11 of 37)" for the next 11 rows, up to 55.

Sensors tab

Sensors shows "Battery", "MCU temp" and any other channels, such as "Humidity ch2". Units follow your Units setting. Only admins see "Position", "GPS altitude" and "Distance". Other logins see only battery and MCU temperature.

Actions tab (admins only)

Other logins see "Admin login needed for actions".

Button	What it does	Confirmation
"Advert"	Sends a zero-hop advert	None
"Flood advert"	Sends an advert across the whole mesh	"Flood advert?" — Send
"Sync clock"	Sets its clock to yours	"Sync repeater clock?" ("Sync room server clock?" on a room) — Sync
"Radio"	Shows its radio settings	None
"Clock"	Shows its clock	None
"Version"	Shows its firmware version	None
" Hold to reboot " (red)	Restarts it	Hold 1.5 s, then "Reboot repeater?" ("Reboot room server?" on a room) — Reboot

"Sync clock" works only when the repeater is 60 seconds or more behind and your clock is set. A repeater's clock can't be moved back.

After a reboot, the toast reads "Reboot sent · log in again in \~30 s", and your session ends.

There is no free-text command line. You can't change the repeater's settings, passwords, radio or power.

The last 3 replies appear under the buttons, for example "3:07 PM Advert → OK - Advert sent". If one reads "No reply · tap to retry", tap it to send again.

Logging out and saved passwords

1. Tap **Log out**.
2. Choose "Log out", or "Log out and forget password".

Logging out only ends the session on your device. The repeater keeps you on its access list until it reboots.

A password is saved only after a successful login with Remember on. A successful login with Remember off deletes it.

Up to 16 repeater passwords are saved; the oldest is dropped when full. A saved password is also forgotten when you remove the contact or factory reset.

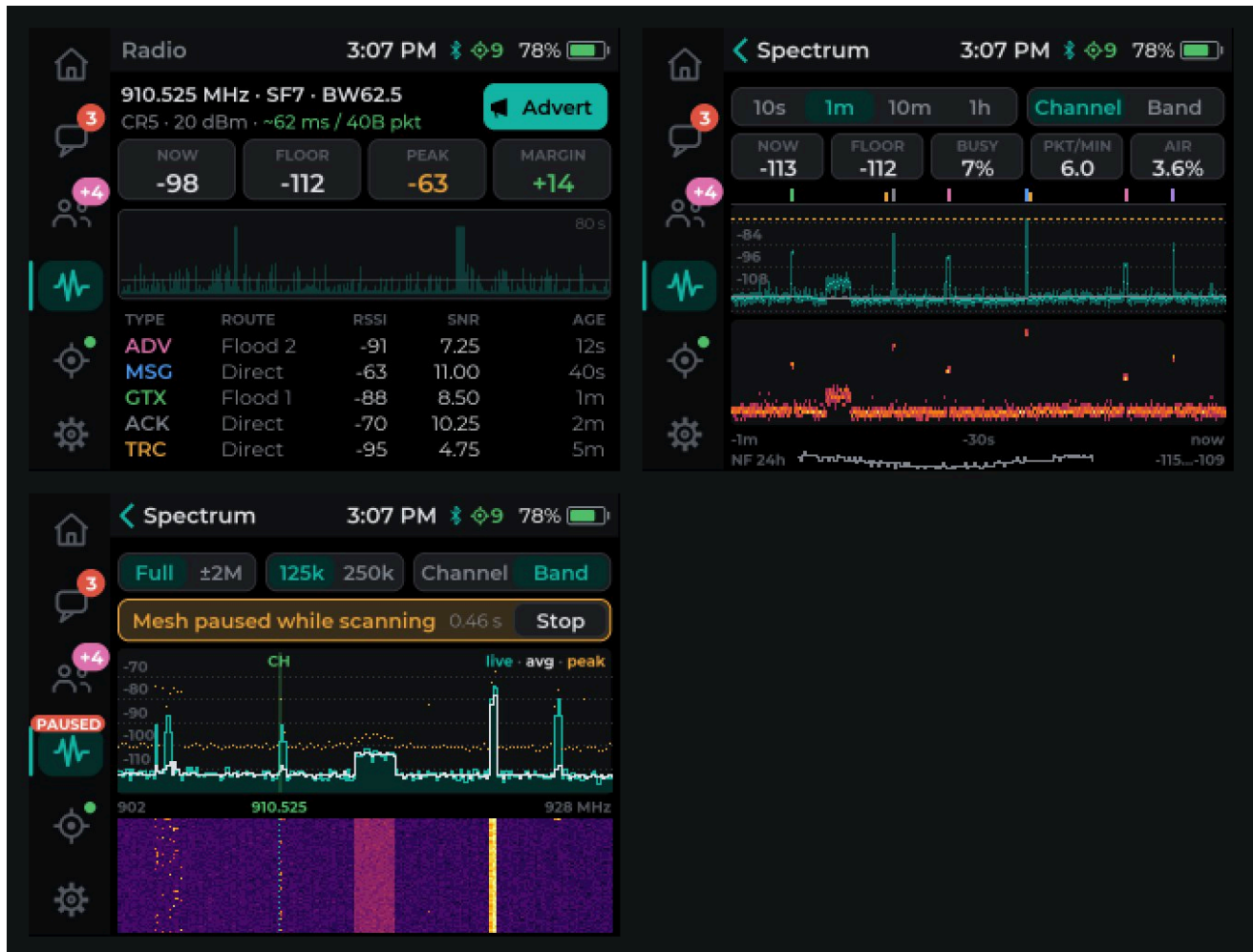
Safety confirmations at a glance

Action	Protection
Clear recents	Hold the trash chip, then confirm
Remove contact	1.5 s hold, then confirm
Reboot repeater or room	1.5 s hold, then confirm
Flood advert	Confirm
Sync clock	Confirm; offered only when the remote clock is behind
Favorite, Reset route, Add repeater, Ping, Scan	No confirmation

07 Radio and spectrum

The Radio page shows your current radio settings, live signal levels and recent packets, and lets you send adverts and inspect the spectrum.

[image: The Radio page, the Spectrum channel view and Band scan]



The Radio page, the Spectrum channel view and Band scan. Sample data.

Open the Radio page

- Tap the radio mast icon on the left rail. It is the fourth icon, after Home, Chats and Nodes.
- Or tap the "RADIO" card on Home.
- If you are already on the Radio page, tap the rail icon again to scroll the packet log back to the top.

The Home "RADIO" card shows the frequency, "SF\ BW\ ", and "NF \ now \ ". When the line is too narrow, it shows "NF \ / \ " instead. A value shows "--" when it is not available.

Rail icon indicators

Small markers above the Radio rail icon show radio activity.

Indicator	Meaning
Orange up-arrow	Flashes briefly when a packet is sent.
Accent-coloured dot (right)	Flashes briefly when a good packet is received.
Amber dot (left of the mast)	The noise floor is not calibrated yet.
Red "PAUSED" pill	A band scan is holding the radio. No other markers show while it is up.

The flashes only show while the screen is lit. They show single packets, not a continuous receive state. The rail is hidden while the keyboard is open, so the "PAUSED" pill is hidden too.

Header

The header shows your modem settings on two lines.

Line	Example	Content
1	"910.525 MHz · SF7 · BW62.5"	Frequency, spreading factor, bandwidth
2	"CR5 · 20 dBm · ~N ms / 40B pkt"	Coding rate, transmit power, estimated airtime

The green "~N ms / 40B pkt" estimates how long a 40-byte packet takes to send. Use it to see the cost of slower presets before you send anything.

The header updates within about 1 second when any of these settings change, from the phone app or from Settings.

The **"Advert"** button (megaphone icon, top right) opens the Send advert sheet.

Meters

Four cards show the current signal conditions. Each card shows "--" until there is data.

Card	Unit	Shows
"NOW"	dBm	Signal level of the latest sample (the newest bar on the graph).
"FLOOR"	dBm	The radio's noise-floor estimate. Shows "~ -110" in dimmer text before calibration.
"PEAK"	dBm	Highest reading in the graph window (amber).
"MARGIN"	dB	NOW minus FLOOR, with a sign, for example "+8". Turns green above 6 dB.

Before calibration, MARGIN uses -110 dBm as the floor. After a change to frequency, bandwidth or SF, FLOOR shows "--" until a new sample arrives.

Signal history graph

The graph shows about 80 seconds of signal level, oldest on the left. The label at top right shows the seconds of history, for example "80 s".

Element	Meaning
Accent bars	Signal level, scaled from 4 dB below the floor to 24 dB above it.
Grey horizontal line	Noise floor

Element	Meaning
Dotted amber line	Peak hold at the tallest bar in the window
"sampling..."	The graph is empty.

Sampling happens only while the Radio or Spectrum page is on screen and the display is on.

- If sampling stops for more than 5 seconds, the graph starts over.
- Changing frequency, bandwidth or SF clears the graph.
- Your own transmissions are never recorded as signal.

Tap the graph to open the full-screen "Spectrum" page.

Packet log

The log lists the last 8 packets received, newest first. It shows "No traffic yet" when empty. Rows cannot be tapped.

Column	Unit	Shows
"TYPE"	—	Packet type code, colour-coded (see below)
"ROUTE"	hops	"Flood \", "Direct \", or "Direct" (no path)
"RSSI"	dBm	Received signal strength
"SNR"	dB	Signal-to-noise ratio, to a quarter dB, for example "7.25"
"AGE"	s / m / h / d	Time since received, for example "12s" or "3m"

The log refreshes about once a second, but not while you are scrolling it. If the clock was set backwards, AGE shows "0s".

Code	Packet type	Colour
REQ	Request	Default text
RSP	Response	Default text
MSG	Direct text message	Blue
ACK	Acknowledgement	Grey
ADV	Node advert	Magenta
GTX	Group or channel text	Green
GDA	Group data	Green
ANQ	Anonymous request	Default text
PTH	Path return	Violet
TRC	Trace	Amber
MPT	Multipart	Default text
CTL	Control	Dim grey
RAW	Raw or custom	Dim grey

Code	Packet type	Colour
?	Unknown type	Default text

Send an advert

An advert announces your node to others. Nothing is sent until you tap **"Send"**.

You can open the "Send advert" sheet in these ways:

- Tap **"Advert"** on the Radio page.
- Long-press the Home "RADIO" card. The sheet opens over Home; you stay on Home.
- Tap the "Advert" tile in the quick panel. The sheet replaces the panel.
- Press a hardware button set to "Send advert".

To send:

1. Choose **"Zero-hop"** or **"Flood"**. The sheet always opens on "Zero-hop".
2. Tap **"Send"**.

Choice	Hint shown	Reach
"Zero-hop" (default)	"Neighbors in direct range only"	Nodes in direct radio range
"Flood"	"Relayed across the mesh (scoped flood)"	Relayed across the mesh. Uses airtime across the whole mesh.

To close without sending, tap **"Cancel"**, tap outside the sheet, or press the User button.

Result	Toast
Zero-hop sent	"Advert sent" (with a tone)
Flood sent	"Flood advert sent" (with a tone)
Failure	"Advert failed"

The advert includes your location unless your advert location policy is set to none.

When you save your Profile in Settings, the device asks "Send advert now?". Choose "Zero-hop (neighbors only)", "Flood (across the mesh)" or "Not now". The same toasts apply.

Spectrum page

Tap the graph on the Radio page to open "Spectrum". Use the back control or swipe back anywhere to return.

The **"Channel" | "Band"** switch at top right picks the view. The page remembers your last choice until the device restarts. After a restart it opens on "Channel".

The Radio page graph keeps sampling while Spectrum is open, so it has no gap when you return.

Channel view

The Channel view shows your current channel over time. It only listens and never pauses the mesh.

The device samples the channel in the background at all times, even with the screen off. Longer histories are ready as soon as you open the page. Changing frequency, bandwidth or SF erases this history.

Time scale

Tap a chip to choose how much time the view covers.

Chip	Window	Refresh	Default
"10s"	10 seconds	10 times a second	Yes
"1m"	1 minute	10 times a second	
"10m"	10 minutes	Once a second	
"1h"	1 hour	Once a second	

Changing the chip clears any cursor or details card. The chip returns to "1m" after a restart.

Stats strip

The five stats update every second. They cover the selected window, but never less than the last minute.

Stat	Unit	Shows
"NOW"	dBm	Latest reading
"FLOOR"	dBm	Noise floor. "\~-110" (dimmed) while uncalibrated; "--" before any reading.
"BUSY"	%	Share of time more than 6 dB above the floor. Amber above 30%, red above 60%.
"PKT/MIN"	packets/min	Packets heard per minute. One decimal below 100, whole number at 100 or above.
"AIR"	%	Airtime used: received packets (estimated) plus your own transmissions (measured).

BUSY shows "--" when no readings were taken with a calibrated floor.

Graph areas

Area	What it shows
Packet lane (thin strip at top)	Each received packet as a marker in its type colour. Width matches its airtime. Short amber ticks mark your transmissions.
Trace graph	Per time slice: a shaded min-max bar, an accent dot for the average and a grey dot for the floor. Dashed line: current floor. Dotted amber line: window peak. Labels every 12 dB.
Density map (below the trace)	How often the signal sat at each level. Dark means rare; bright yellow means most of the time. The scale is colour-blind safe.
Time axis	For example "-1m" · "-30s" · "now". Labels follow the chip.
"NF 24h" sparkline	Noise floor over 24 hours, with its range at right, for example "-118...-112". Shows "collecting..." until data exists.

Full-height grey bars in the trace mark times when you were transmitting. These have no reading. "sampling..." shows when the window is empty.

Inspect a packet

1. Tap on or just below a packet marker in the lane.
2. A details card opens over the trace. The selected marker gets a white outline.
3. The card closes after 6 seconds, or when you tap elsewhere.

Line	Example
1: Type · route · age	"MSG · Flood 2 · 12 s ago"
2: Signal and size	"-97 dBm · SNR 7.25 · 40 B"
3: Who sent it	"from Bob (a3) → me"

Line 3 follows these rules:

- A matched contact shows as name plus hash, for example "Bob (a3)". With no single match, the 2-digit hex shows alone.
- Your own node shows as "me".
- Adverts read "from \" or "from <12 hex characters>".
- Channel packets show the channel name with "#", for example "#public". An unknown channel shows "channel \".
- A flooded packet with at least 1 hop adds "via \".
- "no addresses in this type" shows only when there is no sender and no relay.

Read a point in time

1. Tap the trace or density map. A cursor appears with a readout at top.
2. Tap the same spot again to clear it.

Example readout: "-42 s · min -112 avg -109 max -96 · busy 8%".

- The age reads seconds under 100 s ("-42 s"), minutes with a tenth under 10 minutes ("-1.6 m"), then whole minutes ("-12 m").
- The readout adds "· transmitting" or "· no data" when that applies.
- "busy" is left out when that slice has no calibrated readings.

If the device lacks memory for this view, it shows "Not enough memory".

Band view (band scan)

The Band view sweeps across your region's band and measures signal at each step. Use it to find interference and activity on other frequencies.

The mesh is paused while a scan runs. Your device cannot send or receive messages until the scan stops.

Scan range and step

Chip	Options	Default
Range	"Full" (whole band) or "±2M" (±2 MHz around your channel)	"Full"
Step	"125k" or "250k" (also the measurement bandwidth)	"125k"

"Full" works only on these bands:

- 902–928 MHz (US/CA/AU/NZ/KR/JP)
- 863–870 MHz (EU/IN)
- 433.05–434.79 MHz
- 470–510 MHz (CN)

On any other frequency, only "±2M" is available and "Full" is greyed out. On very wide bands, the device doubles the step automatically, for example 470–510 MHz at "125k" becomes 250 kHz.

You can change chips during a scan. This restarts the averages, peaks and waterfall, and clears the cursor. Your choices are kept until the device restarts.

A Full US band sweep takes about 0.4–0.5 s at "125k" and about 0.2–0.25 s at "250k".

Run a scan

1. Open Spectrum and tap **"Band"**.
2. Choose the range and step chips.
3. Tap **"Start"**. The banner shows "Waiting for radio...".
4. When the radio is idle, scanning begins. The banner turns amber: "Mesh paused while scanning".
5. Tap **"Stop"** when you are done. The radio returns to your channel.

The scan waits until nothing is queued, no ping or remote-admin request is pending, and no packet is being sent or received. If the radio is still busy after 10 seconds, the toast "Radio busy · try again" appears.

State	Banner	Button	Other signs
Idle	"Scanning pauses the mesh"	"Start"	Before the first scan, the plot shows "Sweep \- MHz".
Waiting	"Waiting for radio..."	"Cancel"	Plot shows "Waiting for a quiet radio...".
Scanning	"Mesh paused while scanning" plus sweep time, for example "0.42 s/sweep"	"Stop"	Title adds "• mesh paused". Rail shows red "PAUSED". Plot shows "First sweep..." until the first sweep ends.
Unavailable	"Band scan unavailable" with "Not enough memory" or "This radio can't scan"	None	Chips are disabled.

The screen stays on while scanning.

Read the plot and waterfall

Element	Meaning
Accent filled area	Live sweep
"avg" line	Running average. The first sweep is not averaged.
Dotted amber "peak" line	Peak hold
Green tint labelled "CH"	Your own channel
Gridlines	Every 10 dB
Axis	Low frequency at left, high at right, your channel in green
Waterfall	Past sweeps, newest at top. Brighter means stronger signal. A dotted line marks your channel.

The legend at top right reads "peak · avg · live", or only "live" until averages exist.

- **Tap the plot** to place a cursor, for example "913.0625 MHz · -97 dBm/125k · pk -88 · avg -104". The "pk" and "avg" parts appear once averages exist. Tapping does nothing until a sweep has data.
- **Tap a waterfall row** for that sweep's reading and age, for example "913.0625 MHz · -97 dBm/125k · 3 s ago".
- **Tap the same spot again**, or an empty waterfall row, to clear the cursor.
- **Long-press the plot** to clear the peak hold. The toast is "Peaks cleared".

The "CH" label and legend are hidden while a cursor readout shows.

How a scan stops

A scan never restarts by itself. The radio always returns to your channel.

Cause	Toast
You tap "Stop" or "Cancel"	None
You leave the page, open another page over it, or switch to "Channel"	None
Screen off, lock, hibernate, power notice, firmware update, or an overlay (flashlight, quick panel, sheet, dialog) covers the page	"Scan stopped"
An incoming message takes over the screen	None
3 minutes with no touch	"Scan stopped · 3 min idle"
Frequency, bandwidth, SF or CR changed from the app or Settings	"Scan stopped · radio settings changed"
The mesh needs to send (app message, advert, ping)	"Scan stopped · mesh needs to send"
Repeated measurement failures, or the radio cannot enter scan mode	"Scan error"
Shutdown or restart	None

Problems

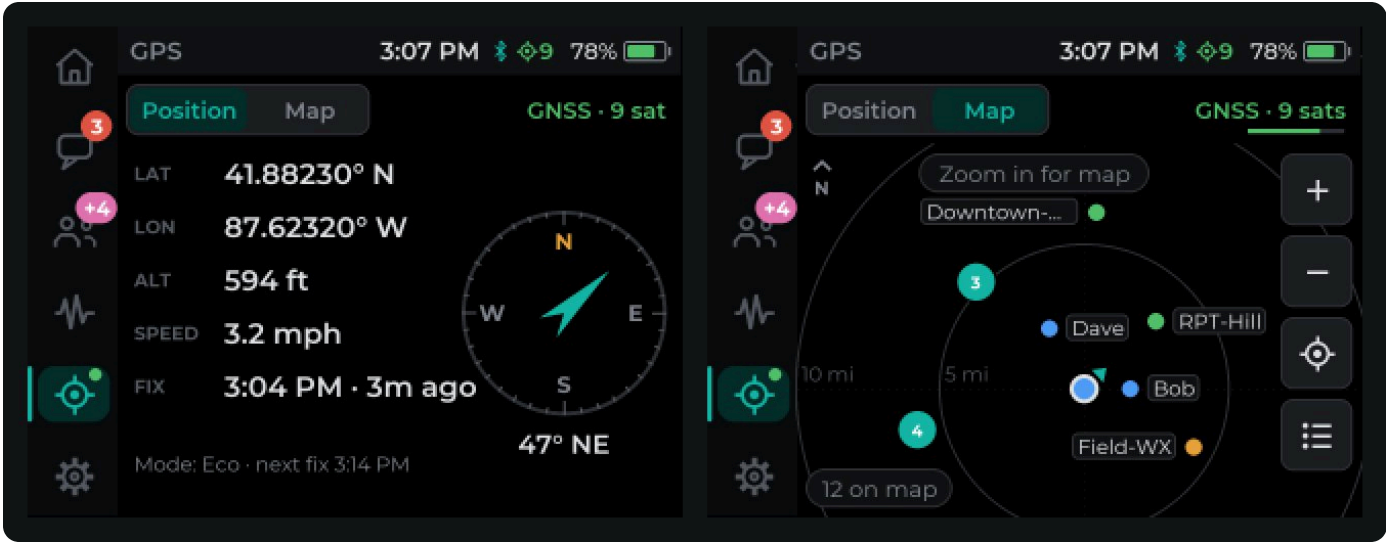
- **"Can't scan this frequency"**: no sweep can be planned for your current frequency.
- **"• radio restoring" in the title**: the radio has not yet returned to your channel. It stays paused so nothing is sent off-channel, and the "PAUSED" pill stays up. Tapping Start shows "Radio still restoring · try again".
- **"Restarting..."**: the radio could not return to your channel after repeated tries. The device restarts and restores your saved radio settings. The restart reason reads "Radio restore failed" (short form "Radio fault").

If your channel changed since your last visit, the old sweep is discarded. A sweep from the same channel stays visible, with waterfall ages such as "2 h ago".

08 GPS and maps

Cairn uses the built-in GPS receiver to show your position, course and speed, and to plot you and other nodes on a map.

[image: GPS position and the map]



GPS position and the map. Sample data.

Where to find GPS

Place	What it does
GPS tab	Has two segments at the top: "Position" and "Map".
Home screen "GPS" card	Tap it to open the GPS tab.
Settings → Location	The "GPS" row sets the GPS mode. The "Map" row chooses a map set.
Quick panel "GPS" tile	Tap it to change the mode. Long-press it to open the GPS settings page.
"Toggle GPS" button action	Switches GPS mode from a hardware button. Its short name is "GPS".
Node detail "Map" button	Opens the map centred on that node.

GPS modes

The GPS receiver uses a lot of power, so you choose how much it runs. The default is "Off".

Mode	What it does	On-screen help
"Off" (default)	The receiver is powered down. The position page and map keep the last fix.	"The receiver is powered down. Position and the map keep the last fix."
"On"	The receiver tracks your position all the time.	"Tracks continuously. The biggest single drain on the battery."
"Eco"	The receiver wakes every 10 minutes for a fix, then sleeps again.	"Wakes every 10 min for a fix, gives up after 90 s without one, then sleeps again."

Change the GPS mode

1. Open **Settings** → **Location** → **GPS**.
2. Under "Mode", tap "Off", "On" or "Eco".

A short toast confirms the change: "GPS on", "GPS off" or "GPS Eco". The mode is saved and kept after a restart. It is a local setting and is not sent over the mesh.

The GPS settings page also has a "Now" card. It updates every second and shows the receiver state:

Text	Meaning
"Fix · N sats" (green)	You have a position. N is the number of satellites used.
"Searching · N in view" (warning colour)	The receiver is looking for a fix. N is the number of satellites in view.
"No GPS data" (warning colour)	The receiver is powered but has sent nothing for over 10 seconds.
"Eco sleep · next fix M:SS"	Eco is asleep. The countdown shows the next wake. It may just read "Eco sleep".
"Off"	GPS mode is Off.
"No receiver" (red)	No GPS receiver was found.

The second line reads "Last fix ... ago" or "No fix yet since boot". The last fix is lost when the device restarts.

How Eco works

- When you choose Eco, it tries for a fix straight away.
- After that, it wakes every 10 minutes.
- Each wake lasts at most 90 seconds. If no fix arrives, the receiver powers down.
- After a fix, the receiver stays on about 15 more seconds so the position can settle. It still powers down by the 90-second limit.

Other ways to change the mode

- **Quick panel "GPS" tile:** shows "On", "Off", "Eco" or "None". Each tap moves On → Eco → Off → On.
- **"Toggle GPS" button action:** switches On to Eco, and Eco or Off to On. It never turns GPS off.

With no receiver, both show the error toast "No GPS receiver". Long-pressing the tile then does the same; it does not open settings.

If the GPS settings page is open when you use the quick panel, the "Mode" control updates to match.

Powering the device off switches the receiver off but keeps your chosen mode. That mode returns at the next start.

GPS status at a glance

Place	What it shows
GPS icon on the left rail	While the receiver is on: a satellite count, green with a fix and grey while searching. In Eco sleep: a small moon. No receiver: nothing.
Quick panel status line	Ends with "N sat", "GPS off", "GPS eco" (asleep) or "no GPS". Example: "78% · ~14h · up 3d 4h · 9 sat".
Settings → Location tile	GPS and map state together, for example "GPS on · map: OKC" or "GPS Eco · no maps".

Place	What it shows
Settings → Location → GPS row	"Off", "On", "Eco" or "No receiver".

"N sats" always counts satellites used in the fix. "N in view" appears while searching. Both counts stop at 99.

The Home "GPS" card updates every 5 seconds:

State	Card shows
No receiver	"None" / "no receiver"
Fix	"Fix" (green), "N sats", then latitude and longitude to 4 decimals, for example "35.1234° N".
Powered, no fix	"No fix" (warning colour). Line 1: satellites in view, prefixed "eco: " in Eco, or "No GPS data". Line 2: "fix Xm ago", "searching..." or "check receiver".
Eco, asleep	"Eco". Line 1: "next 3:40 PM" (clock synced), "next in Nm" (clock not synced) or "sleeping" (no wake scheduled). Line 2: "fix Xm ago" or "no fix yet".
Off	"Off" / "receiver off", or the age of the last fix.

The Position page

The "Position" segment of the GPS tab shows your coordinates, altitude, speed and course.

Status pill

The pill at the top right shows the receiver state:

Pill	Colour	Meaning
"GNSS · N sats"	Green	You have a fix.
"SEARCHING · N in view"	Warning	Looking for a fix.
"NO GPS DATA"	Warning	The receiver is silent.
"ECO · sleeping"	Grey	Eco is asleep.
"OFF"	—	GPS mode is Off.
"NO GPS"	—	No receiver.

While the receiver is on, a thin satellite bar sits under the pill. It is full at 12 satellites.

Data rows

Row	Shows	Format
"LAT"	Latitude	5 decimals plus hemisphere, for example "41.88230° N"
"LON"	Longitude	5 decimals plus hemisphere
"ALT"	Altitude	"ft" (Imperial) or "m" (Metric)
"SPEED"	Ground speed	"mph" or "km/h", 1 decimal
"FIX"	Time of the last fix and its age	For example "12:59 PM · 5m ago" or "12:59 PM · now"

- Units follow **Settings** → **Display** → **Units** ("Imperial" or "Metric").
- With no fix yet, "FIX" shows "searching..." or "No GPS data" while powered, and "no fix yet" otherwise.
- During Eco sleep, the last fix is shown dimmed and "SPEED" shows "--".
- With no data at all, values show "--".

Mode line

The bottom left line shows the mode and what the receiver is doing:

- "Mode: On", "Mode: On · searching" or "Mode: On · no GPS data"
- "Mode: Eco · awake for a fix" or "Mode: Eco · no GPS data"
- "Mode: Eco · fixed, sleeping soon"
- "Mode: Eco · next fix 3:40 PM" or "Mode: Eco · sleeping"

When the page is empty

- **"GPS unavailable" / "No receiver was found at boot"**: the device has no working receiver.
- **"GPS is off" / "Turn it on in GPS settings to see position"**: tap **"GPS settings"** to change the mode. This tab never switches GPS power itself.

The compass

The compass on the right shows your course over ground, not a magnetic heading. It only works while you are moving.

- The ring has tick marks and N, E, S and W, with N highlighted.
- The arrow appears when you move at about 1 mph or faster.
- Below the ring, the readout shows degrees and an 8-point direction, for example "245° SW".
- When you are still or have no fix, the centre reads "no heading", "searching", "no data" or "eco sleep". The readout shows "--".

The map

The "Map" segment shows you and every saved contact that has shared a position. North is always up.

What the map shows

- **Background**: offline map tiles where a microSD map set covers the view. Elsewhere you see a radar view: cross-hairs and labelled range rings around your position.
- **Nodes**: coloured dots, coloured by node type. Nodes close together merge into a bubble with a count. The 10 dots nearest the centre get name labels.
- **Your position**: a filled dot with an outline during a live fix. An arrowhead shows your heading while you move. A hollow grey ring means the last known fix only.
- **North mark** at the top left.
- **Scale bar** at the bottom left, when map tiles are shown. In radar view, the bottom left shows "N on map" instead.
- **Attribution** at the bottom right, when map tiles are shown. Tap it for the full attribution and licence link.

Zoom runs from 2 to 18 without map tiles. Inside a map set, you can zoom 2 levels past the set's most detailed level, up to 19. These extra levels look blocky.

Tiles not yet loaded appear as plain squares with a border. A missing tile shows a grid. Areas outside the set show a grid on the background colour.

If no node has a position, the map reads "No positions yet" / "Nodes that share a location appear here".

Buttons

The buttons run down the right edge, top to bottom. A button is greyed out at the zoom limit.

Button	Action
+	Zoom in one level.
–	Zoom out one level.
Follow (GPS icon)	Recentres on your position and follows it, keeping the zoom. Highlighted while following. With no position, it follows the centre of the nodes.
List	Opens the Nodes tab showing all contacts, sorted by distance. Your usual sort and filter return next time.

Gestures

Gesture	Action
Drag	Pans the map. A quick flick keeps it moving. Panning stops following your position.
Pinch	Zooms with a live preview and a "Zoom N" chip. Works anywhere except the button column.
Double-tap	Zooms in one level on that point.
Tap a dot	Selects the node and shows a card at the bottom. Tap the card to open Node detail .
Tap a bubble	Zooms in 2 levels on the group. At maximum zoom, it lists the nodes instead.
Tap empty map	Clears the selection.
Long-press empty map	Opens the "Map" menu. Long-press does nothing elsewhere on the map.
Tap the GPS tab again	Recentres on your position.

The back swipe does not work on the map, because dragging pans the map.

The selection card shows the node name, then distance and direction with the last-heard age. If some details are missing, it shows only the distance, "heard Xm ago", or the node type. If the node was deleted, tapping the card shows "Node gone".

The bubble list is titled, for example, "12 nodes here (4 shown)". It lists up to 4 names. Tap a name to open Node detail.

Status chips

A chip at the top centre explains what you see.

Chip	Meaning
"No GPS fix – relative view" (warning)	You have no position, so the map is centred on the nodes.
"No map card" (dimmed)	No microSD card is inserted.
"Reading map card...", "Card busy..."	Wait a moment.
"No maps on card"	The card has no map sets.
"Card ejected"	The card was ejected in Storage.
"Card isn't FAT32", "Card not responding", "Map card error", "Low memory for map", "Not enough memory for map" (warning)	The card or map can't be read.

Chip	Meaning
"Zoom in for map", "Zoom out for map"	A map set covers this area at other zoom levels.
"No map for this area"	No map set covers this view.

Tap a card chip for an explanation and a button to retry or reload.

The Map menu

Long-press an empty part of the map to open it.

Item	What it does
"Map card info"	Shows the card type, size and speed, the map sets with their zoom levels, and the card power state. Tap Reload to re-read the card.
"Choose map..."	Lists the first 3 map sets by name, then "Automatic". The current choice is marked with •. Appears only with 2 or more sets.
"Reload card"	Re-reads the card. The toast shows "Reloading map card..." or why it couldn't.

If the card has more than 3 sets, choose the others in **Settings** → **Location** → **Map**. If a set does not cover the current view, you see "That map doesn't cover this view".

Saved view and Node detail

- The map remembers its centre, zoom, follow state and map set when you leave.
- On your first visit, the map zooms to fit your 8 nearest nodes, or about 1 mile with no nodes.
- The Node detail "Map" button centres on that node, selects it, and fits you both in view if you have a position.
- If the node has not shared a position, the button is disabled and shows "No position shared".
- In Node detail, the position row shows "no GPS fix here" when you have no fix. It shows "—" when the node has no position.

The card powers on only while you use the map, the map picker or Storage. When it starts, the screen may pause briefly.

Offline map sets on microSD

Map tiles come only from a microSD card that you prepare on a computer. The device never downloads maps.

Card requirements

- The card must be FAT32. exFAT is not supported.
- An 8–32 GB SDHC card is recommended.
- macOS formats cards over 32 GB as exFAT by default. You can format the card in the device instead (see "Format a card" below).

Prepare map sets

1. On a computer, run the `tools/l2_maptiles.py` tool.
2. Use `build` to make a set from a supported source. Use `estimate` first to check the size.
3. Copy the result to the card's `/l2map/` folder. Each set has its own folder.
4. Insert the card into the device.

Item	Details
Sources	USGS topo (default; US only, up to zoom 16, public domain), Geoapify (free key), Thunderforest, your own MBTiles file, or a local tile server
Refused sources	OpenStreetMap's tile server, OpenTopoMap, CARTO, Esri, Google, Bing, MapTiler Cloud, Stadia
Formats	p8r1e (default, about 15–30 KB per tile) or rgb565 (128 KB per tile)
Filters	night, dim, gray
Set names	a–z, 0–9 and hyphens, 1–24 characters
Sets per card	Up to 8. Extra or badly named sets are skipped.
Other commands	synth (offline test grid), verify, presets, selftest

Map sets are for your personal use only.

Size guide (USGS)

Area around a point	Zoom levels	Approx. size
±5 mi	8–16	33 MB
±10 mi	8–16	126 MB
±25 mi	8–15	200 MB

One screen width covers about 0.4 mi at zoom 16, 1.6 mi at 14, 6 mi at 12 and 25 mi at 10. A good plan is one wide, shallow set plus a detailed set around home.

Choose a map set

1. Open **Settings** → **Location** → **Map**.
2. Tap "Automatic", or tap a set. Each set shows its name and zoom range, for example "name · z8–16".

The toast confirms the choice, for example "Map: Automatic". "Automatic" uses the most detailed set that covers the view. A chosen set is used wherever it covers the view. Elsewhere, Automatic applies.

If your chosen set is missing, "Automatic" shows "Saved map isn't on this card". The last row, **"Storage >"**, opens the Storage page.

If no sets are usable, the page shows one of these messages:

Message	What to do
"Insert a microSD card with maps"	Insert a prepared card.
"Card ejected · Re-scan it in Storage"	Tap Re-scan in Storage.
"No maps on this card (tools/l2_maptiles.py)"	Add map sets to the card.
"Can't read maps: \"	Check the card is FAT32 and fully inserted.
"Card busy...", "Reading card..."	Wait.
"microSD isn't available on this build"	This firmware does not support microSD.

The device notices when you swap cards or change a set on a computer. It retries a card that is not responding after 30 seconds, then at growing intervals up to 8 minutes. A card that isn't FAT32 is retried when you change it or tap Reload.

The Storage page

Open **Settings** → **System** → **Storage** to check the card, remove it safely, delete map sets or format it.

Line	Shows
Header	Card type and size, for example "SDHC · 29.7 GB", or "No card"
State	"Mounted" (green), "Ejected" ("Safe to remove the card"), "Empty slot", "Busy...", "Reading..." or an error
"Space"	"X used · Y free" with a bar. The bar turns to the warning colour at 90% full.
"Format"	For example "FAT32 · 32 KB clusters". An unusable card shows, for example, "exFAT · needs FAT32", or "FAT32 · damaged?".
"Bus"	Card speed, for example "20 MHz · 1-bit · SD32G"

Button	Action
"Eject"	Stops using the card and cuts its power. The toast says "Safe to remove card".
"Re-scan"	Re-reads the card and its map sets. Use it after Eject if you left the card in.

Both buttons are unavailable while a map set is being deleted.

The "MAP SETS" section lists "Automatic" and each set with its size. Tap a row to choose that set.

Delete a map set

1. On the Storage page, find the set under "MAP SETS".
2. Hold its trash icon for 1.5 seconds.
3. In "Delete this map?", tap **Delete**.

The delete runs in the background, even if you leave the page or the screen turns off. A progress row shows "N of M removed". A dot appears on the Settings icon until it finishes. The toast then says "Map deleted".

If you delete your chosen set, the choice returns to "Automatic". If you remove the card mid-delete, the device finishes the job the next time you open Storage.

Format a card

Formatting erases everything on the card and makes it FAT32 with the label "L2MAP".

1. On the Storage page, go to "FORMAT CARD".
2. Hold **"Hold to format card"** for 2 seconds.
3. In "Erase this card?", tap **Erase**.
4. Wait. Do not remove the card or power off. Sound and the mesh pause until formatting ends.

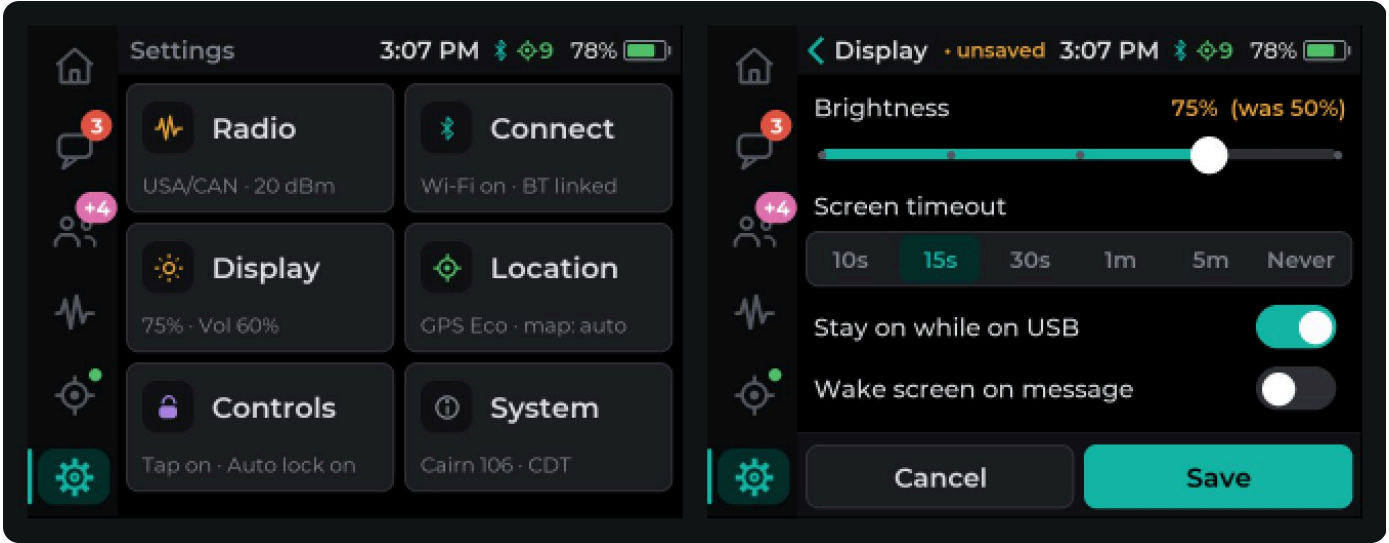
Formatting a 32 GB card takes about 3–6 seconds. The toast then says, for example, "Card formatted · FAT32 · 4 s". If you see "Format canceled", start again and confirm quickly. Formatting is refused while a delete is running.

A hold on the trash or format button is cancelled if your finger moves.

09 Settings

Settings holds every device option in six categories: Radio, Connect, Display, Location, Controls and System.

[image: The six Settings categories and the Display page]



The six Settings categories and the Display page. Sample data.

The Settings home screen

The six tiles are arranged in two columns. Each tile shows an icon, its name and a one-line summary. Tap a tile to open it, or press and hold it. Summaries refresh about once a second.

Tile	Summary shows	Example
Radio	Region and transmit power, or "Custom" if the radio matches no preset	"USA/CAN · 20 dBm"
Connect	Wi-Fi state and Bluetooth state	"Wi-Fi on · BT linked"
Display	Brightness and sound	"75% · Vol 60%" or "75% · Muted"
Location	GPS mode and map choice	"GPS on · map: auto"
Controls	Tap to wake and Auto lock	"Tap on · Auto lock off"
System	Build name and time zone (only the build name if both don't fit)	"Cairn 105 · EDT"

Each category opens a page of cards. Every card shows a live value. Tap a card to open its page, or press and hold it.

Category	Cards
Radio	"Profile", "Channels", "Radio"
Connect	"Bluetooth", "Wi-Fi"
Display	"Display", "Sound"
Location	"GPS", "Map"
Controls	"Buttons", "Lock" (opens "Wake & lock")

Category	Cards
System	"About", "Clock", "Storage", "Firmware", "Restart" (opens "Maintenance"), "Hibernate"

Card values are colour-coded:

- **Green** means the item is working.
- **Amber** means something needs your attention, for example "Card error", or "Last: \ " after a crash.
- **Grey** means the item is absent or empty.

Saving changes

Most option pages have a "Cancel" and "Save" footer.

- "Save" stays greyed out until you change something.
- When you have unsaved changes, the title bar shows "• unsaved" in amber.
- "Cancel" discards your changes and closes the page. It also restores brightness and volume to their saved levels if you were previewing new ones.
- If you leave with unsaved changes, the device asks "Discard changes?". Choose "Keep editing" or "Discard".
- If an incoming message takes over the screen, your unsaved changes are discarded without asking.
- Some changes ask you to confirm with "Cancel" / "Apply", for example "Apply radio change?".
- If a change is not valid, "Save" shows a message explaining why, and the page stays open.
- For [-] value [+] controls, hold a button to repeat it.

The quick panel can change Bluetooth, GPS, mute or volume while one of these pages is open. In that case, a value you have not touched on the page updates to match. Saving the page does not undo the quick panel's change.

Radio category

Profile

Option	Values	Default	What it does
"Node name"	Up to 31 bytes, emoji allowed	—	Sets your name on the mesh, used in adverts and channel posts. Tap it to open the keyboard.
"Share position in adverts"	On / Off	—	Adds your location to each advert.

- A counter ("n/31") shows how many bytes you have used.
- Spaces at either end of the name are trimmed. You cannot save an empty name ("Name can't be empty").
- With position sharing on, adverts carry only 23 bytes of the name. If your name is longer, an amber warning appears: "With position on, adverts carry only 23 bytes".
- Channel posts start with "name: ", which uses part of the space for text. The page shows how many bytes are left.
- Your Bluetooth name changes only after a restart.
- After "Profile saved", the device asks "Send advert now?". Choose "Zero-hop (neighbors only)", "Flood (across the mesh)" or "Not now".

Channels

The list page is titled "Channels · N". Each row shows the channel's name, its type, how many messages it has and an unread count. Group icons are coloured by type: blue for Public, the accent colour for Hashtag and violet for Private.

To join a channel:

1. Tap "+ Join a channel". The right side shows how many slots are in use ("N of M slots").
2. Choose "Hashtag", "Private" or "Public". The default is "Hashtag".
3. Fill in the fields for that type (see the table).
4. Tap "Join". The device shows "Joined \". Nothing is sent over the air.

Type	Fields	Who can read it
"Hashtag"	"Name" (the text after #, no spaces). Capitals matter: #Test and #test are different channels. The key is generated from the name.	Anyone who knows the name
"Private"	"Name" (what this device calls the channel, up to 31 bytes), and "Key" (32 hex digits or base64). Tap "New key" to create a random key.	Only people with the key
"Public"	No fields. Use this to rejoin Public if you left it.	Everyone on the mesh

"Join" is refused with one of these messages: "Give the channel a name", "Type the key, or tap New key", "Key: 32 hex digits or base64", "An all-zero key is not a channel", "Already joined as \" or "All N channel slots are in use".

Channel detail page. Tap a channel to see its type, slot number, who can read it and how many messages are stored. On a Private channel, tap the key card to show or hide the key. "Open chat" opens the conversation.

To leave a channel:

1. Hold "Hold to leave channel" for 1.5 seconds.
2. Choose "Leave" in the "Leave \?" dialog. The default choice is "No".

Stored messages stay in Chats. To rejoin a Private channel, you will need its key again, because the device does not keep it.

Radio (region and transmit power)

Option	Values	Default	What it does
"Transmit power"	−9 to 22 dBm	—	Sets your transmit power. A lower setting saves battery but shortens range. The approximate power in milliwatts is shown next to it.
"Region"	"USA/CAN", "EU/UK 869", "EU classic"	—	Selects a radio preset. The current preset is marked "Current".

Keep transmit power within the limit for your region and for the board. Seeed rates the Wio Tracker L2 at 20 dBm on US915.

Preset	Frequency	Bandwidth	SF	CR
"USA/CAN"	910.525 MHz	62.5 kHz	7	5
"EU/UK 869"	869.618 MHz	62.5 kHz	8	5
"EU classic"	869.525 MHz	250 kHz	11	5

- The "Airtime" line estimates how long a 40-byte message takes to send.
- You cannot set a custom frequency, bandwidth, SF or CR on the device. If the app set a custom radio, the page shows "Now: Custom · ..." in amber. Picking a preset replaces those custom settings.
- "Save" always asks "Apply radio change?" and shows the old and new settings. Tap "Apply" to confirm.
- If client repeat is on and the new frequency does not allow it, "Save" is refused with "Disable repeat first".

Connect category

Bluetooth

Option	Values	Default	What it does
"Bluetooth"	On / Off	On	Lets the app connect. The setting is kept after a restart. It cannot be changed during update mode.
"Use auto"	Button	Automatic PIN	Picks a new random pairing PIN at every boot. Read the PIN on this page when you pair.
"Change PIN..."	Six digits	—	Sets a fixed PIN that stays the same across restarts. Entering 000000 switches back to an automatic PIN.

- The info card shows the app link as "Connected", "Not connected" or "Bluetooth off". It also shows the "Pairing PIN" and whether it is "auto · new each boot" or "fixed · kept across boots".
- "Use auto" and the PIN pad (titled "BLE PIN") save immediately. You do not need the page's "Save".

Wi-Fi

Option	Values	Default	What it does
"Wi-Fi"	On / Off	—	Turns Wi-Fi on. Leave it off to save battery.
Network card	Tap to open "Networks"	—	Shows the network name and connection status.
"Set clock from internet"	On / Off	—	Sets the clock from internet time servers. The hint shows when the clock last synced.
"App over Wi-Fi"	On / Off	—	Lets the MeshCore app connect over Wi-Fi on port 5000.
"Trusted devices"	Opens a list	None	Lists the phones that can connect without asking.
"Saved networks"	Opens a list	None	Lists the networks this device remembers.

- Wi-Fi uses about 20–30 mA more power while connected. Bluetooth stays on.
- Status messages shown in red include "Wrong password", "Network not found", "Could not connect" and "Not enough memory for Wi-Fi".
- Turning off Wi-Fi or "App over Wi-Fi" while the app is linked asks "Apply Wi-Fi change?". The app will disconnect.
- Turning off Wi-Fi during update mode also asks for confirmation. Update mode stops and the device restarts.

To join a network:

1. Tap the network card to open "Networks". The device scans automatically. Tap "Rescan" to scan again.
2. Tap a network: - **Saved network:** the device joins it directly. - **Open network:** confirm with "Join". - **Secured network:** type the password (8–63 characters) and tap "Join". The password is visible while you type.
3. When the device shows "Connected to \"", it returns to the Wi-Fi page. If you see "Wrong password", the keyboard reopens with your text kept.

To join a hidden network: tap "Hidden network..." at the bottom of the list. This row appears after a scan finishes. Type the network name exactly as it is set on the router, then tap "Next". Type the password, or leave it empty for an open network.

Saved networks. The device saves up to 8 networks that gave it an address, most recent first. When the list is full, the network you connected to least recently is dropped. Wi-Fi reconnects to the strongest saved network in range.

- Each row shows "Connected now", "Connecting..." or "Last connected \".
- Tap a row to choose "Connect now" or "Forget". Forgetting a network erases its saved password.
- "Forget all" erases every saved network.
- This page works with Wi-Fi off. You cannot use it while the device is locked.

App access over Wi-Fi. When the app connects over Wi-Fi, the device asks "Allow app over Wi-Fi?". Allow only your own phone.

Choice	Result
"Cancel" (default)	Refuses the connection. That address is refused for 2 minutes.
"Allow once"	Allows this connection only.
"Always allow"	Adds the phone to Trusted devices, so it connects without asking. This option does not appear when connecting through range extenders, port forwarding or VPNs.

- An unanswered prompt closes after 60 seconds.
- If the screen locks or a message takes over before you answer, the prompt appears again.
- You can trust up to 4 devices. Trusting a fifth replaces the device seen least recently.
- To remove a trusted device, hold its trash icon for 1.5 seconds and confirm with "Forget".
- To remove all of them, hold "Hold to forget all" and confirm with "Forget all". A linked app disconnects immediately.

Display category

Display

Option	Values	Default	What it does
"Brightness"	10–100 % (snaps to 10, 25, 50, 75 or 100)	75 %	Sets screen brightness. The screen previews the new level while you drag.
"Screen timeout"	"5s", "10s", "15s", "30s", "1m", "5m", "Never"	15s	Sets how long the screen stays on after your last action. If the app set a time not in this list, it is kept unless you change it here.
"Stay on while on USB"	On / Off	Off	Keeps the screen on while USB power is connected.
"Wake screen on message"	On / Off	On	Turns the screen on when a message arrives. When it is off, the Display card shows "no wake" in amber.
"Text size"	"Normal" / "Small"	Normal	Sets the size of message text.
"Compact messages"	On / Off	Off	Shows each message on one line in conversations.
"Units"	"Imperial" / "Metric"	Imperial	Sets the units used for distances and similar values.

Sound

Option	Values	Default	What it does
"Tones"	On / Off	—	Turns speaker alerts on or off. This is the mute switch.

Option	Values	Default	What it does
"Volume"	5–100 % in 5 % steps	60 %	Sets alert volume. A preview plays when you release the slider. To silence alerts, turn "Tones" off.
"Direct message"	Tone list*	Chirp	Plays when a direct message arrives.
"Channel"	Tone list*	Kerplop	Plays when a channel message arrives.
"Advert heard"	Tone list*	Advert	Plays when an advert is heard. Choose "None" to turn off advert tones.
"Sent / confirm"	Tone list*	Tick	Plays when a message is sent or confirmed.
"Startup sound"	"None", "Chime", "Soft", "Classic"	Chime	Plays when the device starts.

*Tone list: "None", "Chirp", "Kerplop", "Tick", "Advert", "Ding", "Double", "Rising", "Falling", "Bell", "Trill".

- Use < and > to change a tone. Each tone plays as you select it.
- "Test" plays the message tone. It is disabled while the device is muted.
- Previews and "Test" play at the volume you have set on the page, even before you save.
- To mute at any time, triple-click User or use the quick panel.

Location category

GPS

Mode	What it does
"Off" (default)	Powers down the receiver. Your position and the map keep the last fix.
"On"	Tracks your position continuously. This is the biggest single drain on the battery.
"Eco"	Wakes every 10 minutes to get a fix. If it finds none within 90 seconds, it sleeps again.

The "Now" card shows the GPS status:

Status	Colour	Meaning
"Fix · N sats"	Green	The GPS has a position.
"Searching · N in view"	Amber	The GPS is looking for a fix.
"No GPS data"	Amber	The receiver is sending no data.
"Eco sleep · next fix m:ss"	Plain	In Eco mode, sleeping until the next fix attempt.
"Off"	Grey	GPS is off.
"No receiver"	Red	No GPS receiver was found.

The card also shows "Last fix \ ago" or "No fix yet since boot".

Map

Option	Values	Default	What it does
Map set	"Automatic" or one map set from the card	Automatic	Selects which map set to use. "Automatic" picks the best map for the current view.

- Tap a row to choose it. Your choice is saved immediately, with no "Save" button.
- Each map set row shows its title and its zoom range ("z\-\").
- If your chosen set is missing from the card, the page shows "Saved map isn't on this card".
- Card messages include "Insert a microSD card with maps", "Card ejected · Re-scan it in Storage" and "No maps on this card (tools/l2_maptiles.py)".
- "Storage ›" opens the Storage page.

Controls category

Buttons

Option	Values	Default	What it does
"Click" (press once)	Action list*	"Quick panel"	Runs the chosen action.
"Double-click" (press twice)	Action list*	"New message"	Runs the chosen action.
"Long press" (hold 1 s)	Action list*	"Back"	Runs the chosen action.

*Actions:

Action	What it does
"None"	Does nothing, and does not close an open sheet.
"Quick panel"	Opens the quick panel. Press again to close it.
"Back"	Goes back one screen.
"New message"	Opens the recipient picker for a new message.
"Send advert"	Lets you choose zero-hop or flood. Nothing is sent until you tap a choice.
"Ping last node"	Asks "Ping \?" for the last node you heard from or messaged.
"Scan repeaters"	Opens Nodes › Scan and starts a scan.
"Toggle GPS"	Turns GPS on from Off, then switches between On and Eco.
"Flashlight"	Turns the screen full white until you press a button or touch the screen.

Some button rules cannot be changed:

- A triple-click always mutes.
- While the device is locked, a click or long press only lights the lock screen. A double-click or triple-click unlocks it.
- If the screen is dark, the first press only wakes it.
- If the flashlight is on, any press only turns it off.
- If a sheet, dialog or panel is open, a press closes it.
- A long press while the startup screen is showing starts the USB rescue console.

Wake & lock

Option	Values	Default	What it does
"Tap to wake"	On / Off	On	A touch lights the dark screen. The touch does nothing else.
"Auto lock"	On / Off	Off	Locks the device when the screen times out. Turning it off also unlocks the device.
"Touch unlock"	On / Off	Off	Lets you unlock by holding the padlock for 1.5 seconds.

To unlock, double-press WAKE or User. To lock immediately, hold WAKE for 1 second.

System category

About

This page is read-only and scrolls.

Section	Shows
Identity	Build number, node name, and how the firmware was installed (e.g. "app1 · OTA")
Version	"MeshCore \" and the compile date
"MESH"	"Public key" (tap it to see the full 32-byte key), the number of contacts and the number of channels
"HEALTH"	Uptime, CPU, battery percentage and voltage, and free memory
"LAST RESTART"	The reason for the last restart, how long ago it was, and the watchdog state

Clock

Option	Values	Default	What it does
"Style"	"12h", "24h", "Analog"	12h	Sets the clock format.
"Time zone"	"Eastern", "Central", "Mountain", "Arizona", "Pacific", "Alaska", "Hawaii", "Custom"	Custom, UTC+0	Sets your time zone. Daylight saving is applied automatically.
"UTC offset" (Custom only)	UTC−12:00 to UTC+14:00 in 15-minute steps	See note	Sets a fixed offset with no daylight saving. The values wrap around at each end.

Zone	Standard offset	Daylight saving
Eastern	UTC−5	Yes
Central	UTC−6	Yes
Mountain	UTC−7	Yes
Arizona	UTC−7	No
Pacific	UTC−8	Yes
Alaska	UTC−9	Yes
Hawaii	UTC−10	No

- The "Daylight saving" row shows this year's dates and "On now" or "Off now". Arizona and Hawaii show "Not observed: the clock never shifts".
- The zone abbreviation (e.g. "EDT") and "On now" or "Off now" appear only after the clock is set. Until then, only the UTC offset is shown.
- If Custom is already saved, the "UTC offset" control starts from your saved offset. Otherwise, it starts from your current zone's offset right now, for example UTC−4 for Eastern in summer.
- The "Now" card previews the time and date. If the clock is not set, it shows "--:--" and "Not set: syncs from the app or GPS".

Storage

The card info shows the card type and size, its state, the used and free space, its format and its bus speed.

Control	What it does
"Eject"	Prepares the card for removal. Remove the card when you see "Safe to remove card".
"Re-scan"	Reads the card again, for example after you reinsert it.
"MAP SETS" trash (hold 1.5 s)	Deletes a map set after you confirm with "Delete". This cannot be undone.
"Hold to format card" (hold 2 s)	Erases the card and formats it as FAT32 (label L2MAP) after you confirm with "Erase".

- Maps can be read only from FAT32 cards. Format exFAT cards before you use them for maps.
- During formatting, do not remove the card or power off. Sound and the mesh pause until formatting finishes.
- Formatting is canceled if you leave the page or the device locks. It is also canceled if formatting does not start within 5 seconds of confirming.
- "Eject" and "Re-scan" wait while a delete is running.

Firmware update

Control	What it does
Online update card	Checks GitHub for new releases once a day while on Wi-Fi. Tap it to check now, or to open a release that was found.
"Install" (on a release page)	Downloads the release from GitHub and installs it. Bluetooth is off until the device restarts.
Wi-Fi card	Shows the network and its status. Tap it to open Wi-Fi settings.
"Start update mode"	Lets a computer upload firmware through a browser or PlatformIO. Bluetooth turns off, and the device restarts afterwards. Needs Wi-Fi.
"New password"	Creates a new update-mode password. The old password stops working immediately.
"Stop"	Ends update mode.

To update from a computer:

1. Connect to Wi-Fi, then tap "Start update mode" and confirm with "Start".
2. On the computer, open the URL shown on the device. Sign in with the "User" and "Password" shown on the device.
3. Choose the "-update.bin" file and tap "Install". Keep the device powered.

Anyone with the password can install firmware. Update mode stops by itself after 10 minutes without activity, and after 30 minutes at most.

Maintenance (Restart card)

Option	Values	Default	What it does
"Restart"	Button	—	Restarts the device. The mesh and Bluetooth drop for about 10 seconds. Nothing is lost.
"Reset UI settings"	Button	—	Resets units, tones, volume, buttons, lock touches and the map view. Radio settings, contacts, chats and Wi-Fi are kept.
"Keep across restarts"	On / Off	On	Saves chat history so it survives a restart. Turning it off erases the saved copy immediately.
"Hold to clear chat history"	Hold 1.5 s	—	Deletes all stored messages and the saved copy after you confirm with "Clear".
"Hold to factory reset"	Hold 2 s	—	Erases contacts, channels, chats, this node's identity, Wi-Fi and all settings, then restarts. Maps on the card are kept.

- The top line shows the reason for the last restart. It is amber after a fault.
- Scrolling cancels a hold that is in progress.

Hibernate

"Hold to hibernate" (hold 1.5 s, then confirm with "Yes") powers the device off until you press User.

- While hibernating, nothing is received or relayed.
- Settings are kept, including the GPS mode.
- Chats are kept only if "Keep across restarts" is on in Maintenance.

10 Power and battery

This chapter covers how to read the battery level, charge the device, make the battery last longer, and put the device into hibernation.

Battery icon in the status bar

The battery icon is at the top right of the status bar. It shows the charge percent inside the outline, with no "%" sign.

Icon state	Meaning
Green fill	Normal charge level.
Amber fill	20% or below, on battery or with an unknown power source.
Red fill	10% or below, on battery or with an unknown power source.
Lightning bolt at the right end	The status is "Charging".
USB plug at the right end	The device is on USB power, or the status is "Full".
Solid green fill reading "100" with a plug	Full and still on the charger. On battery alone, 100% never looks like this.
Empty grey outline reading "--"	No battery reading is available.

Tap the clock, or anything to the right of it, to open the quick panel. The quick panel status line also shows the battery, for example "78% · ~14h · up 3d 4h · 9 sat". On external power it shows "78% · USB ..." instead.

Home "BATTERY" card

The Home screen has a "BATTERY" card that summarizes the battery.

Item	What it shows
Large percent	Turns amber at 20% or below when not on external power. This includes "Source unknown".
Small value beside it	On USB: "chg" (accent colour) or "full" (green). On battery: "~Nh", the estimated hours left, once an estimate exists.
First line	Voltage and source, for example "4.01 V · USB" or "4.01 V · BATT". Shows only the voltage if the source is unknown.
Second line	A small trend graph. It reads "logging..." until it has two samples.

Tap the card to open the "Battery" page.

You can also see the battery in two other places:

- **About** → **HEALTH** → **"Battery"**: for example "78% · 4.01 V". On external power it adds " · Charging", " · Full" or " · USB".
- **Boot screen**: the "BATTERY" column shows an icon and "N%". The icon is green above 20% and amber at 20% or below. A bolt appears on external power. A dash means there is no reading yet.

The Battery page

The "Battery" page shows the battery level, the power source, time remaining and a discharge graph.

Top section

- **Large percent:** red at 10% or below and amber at 20% or below. These colours apply only when not on USB.
- **Voltage:** to three decimals, for example "4.012 V".
- **Source line:** shows one of these:

Source line	Colour	Meaning
"Charging"	Accent	On USB, and the battery appears to be charging.
"Full"	Green	On USB, and the battery appears full.
"USB power"	Accent	On USB, and the charge state is unclear.
"BATT \~Nh"	Green	On battery, with about N hours remaining.
"On battery"	Secondary text	On battery, with no estimate yet.
"Source unknown"	Secondary text	The device could not tell whether USB is connected.

Time remaining

The estimate appears once the battery has dropped at least 2% over at least 10 minutes. It assumes the battery keeps draining at that rate. The highest value it shows is 999 h.

- A full battery at "100%" shows no estimate until the percent starts to drop.
- Plugging in or unplugging USB starts the estimate over.

Discharge graph

- The vertical axis is marked "100%", "50%" and "0%". A dashed amber line marks 20%.
- The line is accent-coloured on USB, amber when the battery is low, and green otherwise.
- The horizontal axis runs from the start of the session (for example "-1h 5m") on the left to "now" on the right.
- It shows "logging..." until it has two samples.

The graph always covers the whole session. As the session gets longer, the samples are spaced further apart. The graph starts over when you plug in or unplug USB. It also starts over after every restart.

Stat cards

Card	What it shows
"UPTIME"	How long the device has been running.
"CPU"	The processor temperature.
"HISTORY"	The number of graph samples and how far apart they are, for example "63 @ 5m" or "12 @ 20s".

While the screen is on, the page updates every few seconds. While the screen is off, the battery is sampled every 30 seconds.

Charging

To charge, connect a USB-C cable to a wall charger or a computer. Both count as USB power.

The device cannot read the charger's status directly. "Charging" and "Full" are estimates based on the battery voltage.

Status	When it shows
"Charging"	The voltage has been rising for about 6–7 minutes, or it is below 4.15 V. Once shown, it stays up to 4.16 V.
"Full"	The voltage is 4.18 V or higher and not rising. Once shown, it stays down to 4.17 V.
"USB power"	The voltage is between these levels, or there is no battery reading.

Things to expect while charging:

- The percent reads higher than the true level while charging. Charging current raises the voltage.
- After you plug in, it takes about 6–7 minutes for the device to detect a rising voltage.
- The status can say "Full" while the last few percent are still charging.
- The status can go back to "Charging" after charging has finished, if the battery voltage drops below 4.15 V.

The percent is based on the voltage curve for the 3000 mAh battery. 4.18 V is 100%, 3.80 V is 70% and 3.30 V is 0%. This curve has not yet been checked against a full discharge on this device.

Low-battery warnings

The device warns you twice as the battery runs down.

Level	Message	Shown for
20%	"Battery N% · charge soon"	3 s
10%	"Battery N% · charge now" (error style)	4 s

- If there is no voltage reading, the messages say "Battery low · charge soon" or "Battery low · charge now" instead of a percent.
- Each warning plays a falling tone at your saved volume. The tone is silent when the device is muted.
- Each warning appears only once each time the battery runs down. Connecting USB resets both warnings.
- If you turn the device on below 10%, you get only the 10% warning.
- If the screen is off, the tone plays right away. The message appears the next time the screen turns on.

Automatic power-off when the battery is empty

The device turns itself off before the battery is fully drained. This happens when the battery stays at 0% (below 3.30 V) for about 30 seconds while on battery.

- It never happens on USB power, or when the device cannot tell whether USB is connected.
- The screen turns on and shows "Battery empty — powering off" / "Charge, then press User to power on".
- The device turns off after you release all buttons and lift your finger from the screen. If you keep holding, it turns off after 10 seconds anyway.
- If you plug in USB first, power-off is cancelled. You see "Power connected · staying on", and both warnings are reset.

To turn the device back on, charge it and press **User**. On the "Restart" card, the last restart reason then reads "Low battery". About shows "Woke (low battery)".

Hibernate

Hibernate turns the device off until you press **User**. Use it to save the battery when you are not using the device.

While the device hibernates, it does not receive or relay any messages. The radio, GPS, screen, touch, SD card and Grove port are all switched off.

What is kept

"Keep across restarts" (Maintenance page)	Settings, including GPS mode	Chats
On	Kept	Kept
Off	Kept	Cleared

Before turning off, the device also saves the map view, pending contact edits and any queued outgoing messages.

To hibernate

1. Open **Settings** → **System**, then tap the "Hibernate" card. Its summary reads "Power off".
2. Read the page headed "Power off until User is pressed".
3. Press and hold the red "Hold to hibernate" button for 1.5 seconds.
4. In the "Hibernate?" dialog, tap **Yes**. **No** is selected by default.
5. When "hibernating..." / "Press User to power back on" appears, lift your finger and release all buttons.

The device turns off once you have let go. It waits as long as you keep holding. If a message arrives before it turns off, hibernation is cancelled.

To wake the device

Press the **User** button.

The side **WAKE** button cannot wake the device from hibernation. Waking is a full restart, including the boot screen. Afterwards, the "Restart" card reads "Last: Hibernate" and About reads "Woke from hibernate". In some cases About may read "Woke from sleep" instead.

The power switch

The slide switch on the right edge is the board's hardware power switch. Switching it off cuts power at once, without the saves that Hibernate makes first. Changes from the last few seconds, such as recent messages or contact edits, can be lost. To turn the device off cleanly, use Hibernate.

Saving power with GPS

GPS is the biggest single drain on the battery. Set the GPS mode under **Settings** → **Location** → **GPS**.

Mode	What it does	Default
"Off"	Turns the receiver off. Your position and the map keep the last fix.	Yes
"On"	Tracks your position continuously. Uses the most power.	
"Eco"	Wakes every 10 minutes for a fix. If it gets no fix within 90 seconds, it gives up and sleeps again.	

When you save, the device shows "GPS on", "GPS Eco" or "GPS off".

The "Now" card shows the current GPS state:

"Now" card	Meaning
"Fix · 9 sats"	The device has a position.
"Searching · 12 in view"	The device is looking for a fix.
"No GPS data" (amber)	No data is coming from the receiver.
"Eco sleep · next fix m:ss"	Eco mode is sleeping until the next fix.
"Off"	GPS is off.
"No receiver"	No GPS receiver was found.

A second line reads "Last fix X ago" or "No fix yet since boot".

You can also change the GPS mode in two other ways:

- **Quick panel:** tap the GPS tile.
- **User button:** if you assign the action "Toggle GPS" to it, On switches to Eco, and Eco or Off switches to On.

While Eco is asleep, the GPS cell on the side rail shows a moon icon instead of the satellite count. The quick panel line shows "GPS eco".

Other power-related settings

Transmit power

Open **Settings** → **Radio** → **Radio** and find "Transmit power".

Setting	Range	Notes
"Transmit power"	-9 to 22 dBm	Set with a stepper. A bar runs from "-9" to "22" with the caption "Lower saves battery, shortens range".

The caption shows the approximate power, for example " ≈ 0.3 mW". While a change is unsaved, it shows " $\approx$ N mW · was X dBm" in amber. Keep within your region's limit and the board's rating (20 dBm on US915, per Seeed).

Display

The screen also uses power. Open **Settings** → **Display** → **Display**.

Setting	Options	Default	Confirmation
"Brightness"	Slider, 10% to 100%	75%	"Brightness N%"
"Screen timeout"	5s, 10s, 15s, 30s, 1m, 5m, Never	15s	"Screen off after \", or "Screen stays on" for Never
"Stay on while on USB"	On / Off	Off	"Stays on with USB" / "Times out on USB too"
"Wake screen on message"	On / Off	On	"Wake on message on" / "Wake on message off"

"Stay on while on USB" keeps a lit screen on while on external power. It does not apply while the device is locked, and it does not turn on a dark screen. The flashlight still turns off after its own 10-minute limit.

The quick panel also has "Brightness" and "Volume" sliders.

Automatic power saving

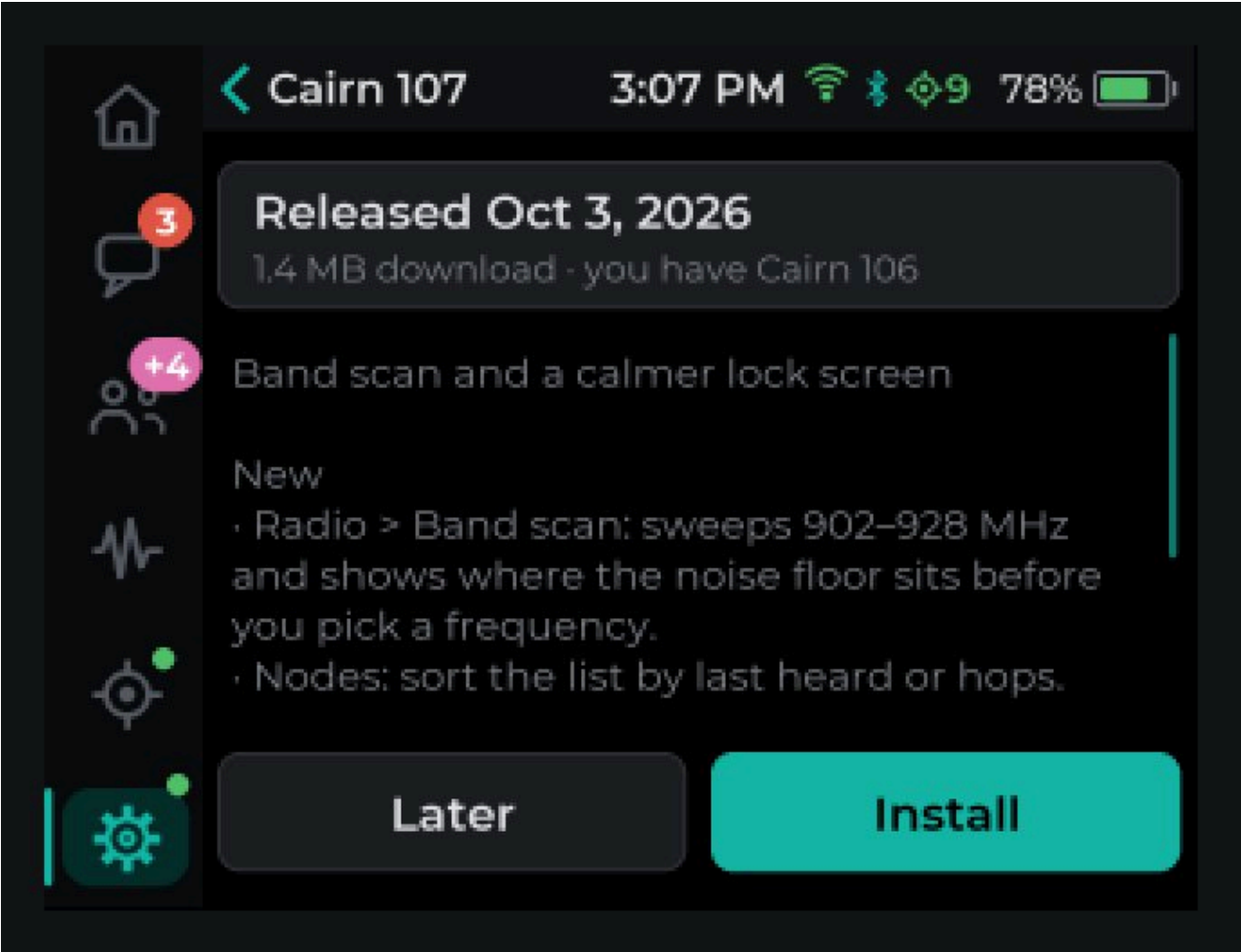
The device saves power on its own, with no setting needed.

- While the screen is off, the processor slows down and the SD card is switched off when idle.
- Bluetooth advertising slows down 30 seconds after the device starts or a phone disconnects.
- If tap-to-wake is off, touch is turned off while the screen is dark. Press a button to wake the screen.

11 Updating and recovery

You can update Cairn over Wi-Fi or USB, and recover a misbehaving device with the Maintenance page or the rescue CLI.

[image: A firmware update offered over Wi-Fi]



A firmware update offered over Wi-Fi. Sample data.

Firmware update page

To open it, go to Settings → System and tap the **"Firmware"** card. The page is titled **"Firmware update"**.

The **"Firmware"** card on the System page shows one of these summaries:

Card text	Meaning
"Ready"	Wi-Fi is connected, so you can start update mode.
"Needs Wi-Fi"	Wi-Fi is not connected.
"Update mode on"	Update mode is running.
"Cairn N available"	A newer release was found online.

Card text	Meaning
"Installing..."	An online install is in progress.

The page has four parts:

- **Firmware card:** the build name (for example "Cairn 105"), a slot chip, "MeshCore \ · \", and either "Flashed over USB" or "Installed over Wi-Fi".
- **Online-check card:** see "Online update from GitHub" below.
- **Wi-Fi card:** shows the Wi-Fi state. Tap it to open Wi-Fi settings.
- **Bottom button:** "Start update mode" when Wi-Fi is connected. Otherwise it reads "Open Wi-Fi settings".

The Wi-Fi card can show these states:

Wi-Fi card	Meaning
"\ " / "Connected · \"	Connected. Update mode is available.
"Wi-Fi off"	Wi-Fi is turned off.
"No network chosen"	No network has been chosen in Wi-Fi settings.
"Not in range · update mode needs Wi-Fi"	The chosen network is out of range.
"Connecting..."	The device is joining the network.
"Wi-Fi low on memory"	Wi-Fi could not get enough memory.
"Wi-Fi not available" / "Not in this build"	This build has no Wi-Fi.

Firmware slots and rollback

The device stores firmware in two slots, app0 and app1. A Wi-Fi update writes to the slot that is not running. The device switches to that slot when it restarts.

The slot chip on the Firmware update page and on About → Firmware shows which slot is running and how its firmware was installed:

Chip	Meaning
"app0 · USB"	Flashed over USB.
"app0 · OTA" or "app1 · OTA"	Installed over Wi-Fi.

New firmware is confirmed after it has run for 30 seconds. If it crashes before then, the device should go back to the previous firmware at the next reset. This rollback has not yet been verified on hardware.

Online update from GitHub

The device checks the public GitHub repository `cvhviz/Cairn` for the latest release. **Nothing installs automatically.**

Note: The online update path has not yet been tested on hardware. If it fails, update from a computer instead (see the next sections).

Automatic check

- The first check runs 2 minutes after Wi-Fi connects. After that, the device checks at most once a day.

- If a check fails, it tries again after an hour.
- Only the latest release counts. Pre-releases are ignored.

When a newer build is found

- A green dot appears on the Settings rail icon.
- The "Firmware" card reads "Cairn N available".
- A toast, "Cairn N available", appears once for each new build. Tap it to open the release.
- The card and the dot stay until you install the build.

Online-check card

The first line reads "Check for updates" or "Cairn N available". The time of the last check appears on the right: a time today, "Yesterday", or a date such as "Sep 30".

Second line	Meaning
"Checks GitHub daily on Wi-Fi"	No check has been run yet.
"Checking..."	A check is running.
"Up to date (Cairn 105)"	You have the latest build.
"Tap for details · 1.4 MB"	A release is available. Its download size is shown.
"No releases yet"	The repository has no releases.
"Cairn N has no Wio L2 file"	The latest release has no file for this device.
An error message	The check failed. See the troubleshooting table.

If a release is available, tapping the card opens it. Otherwise a tap starts a check now. Without Wi-Fi, a tap shows "Connect to Wi-Fi first".

To install a release:

1. Tap the online-check card (or the "Cairn N available" toast). The release page opens with the title "Cairn N".
2. Read "Released \ " / "\ download · you have Cairn \ " and the release notes.
3. Tap **Install**. Tap **Later** to leave it for now.
4. In the "Install Cairn N?" dialog, tap **Install**. The dialog says "Downloads X MB from GitHub. Bluetooth is off until the device restarts."
5. Keep the device powered while it shows these stages: "Preparing...", "Connecting to GitHub...", "Downloading Cairn N" with a progress bar, "Checking the firmware...", and "Installed · restarting..."

You cannot cancel an install, and **Back** does not work while it runs. Bluetooth stays off until the restart.

The release page can also show "Checking GitHub...", "Could not check", "No update", "Connect to Wi-Fi to see this release." or "No update loaded". The bottom button then changes to "Check again" or "Open Wi-Fi settings".

If the install fails, the restart screen shows "Update failed · \ ". The device then restarts on the old firmware. A failed download starts over from the beginning. It does not resume.

Update from a computer over Wi-Fi ("update mode")

Update mode lets a browser or PlatformIO on the same network upload a firmware file to the device.

Before you start

- The device must be on Wi-Fi.

- Use the `-update.bin` file (or `firmware.bin`) from a Wio L2 **touch** build. The device refuses older keypad builds, Meshtastic images and files for other boards.
- Bluetooth turns off for the whole session, so a phone app on Bluetooth disconnects. The app link over Wi-Fi keeps working.

To start update mode:

1. On the Firmware update page, tap **Start update mode**.
2. In the "Start update mode?" dialog, tap **Start**. The dialog says "A computer uploads the .bin. Anyone with the password can install. Bluetooth off; restarts after."
3. The ready screen shows: - "Open in a browser" with the address `http://<ip>/` - "User": `meshcore` - "Password": 12 characters, made of lowercase letters and digits. The letters `i`, `l` and `o` and the digit `1` are never used. - "PlatformIO: `espot -i \ -a \`" - "Stops by itself in `m:ss`"

A new password is created for every session. It is not saved on the device. To replace it, tap **New password**, then **Change**. The old password stops working at once.

To upload from a browser:

1. On the computer, open `http://<ip>/`.
2. When the browser shows "Password needed (shown on the device)", sign in with user `meshcore` and the password from the device.
3. On the "Firmware update" page in the browser, choose the `.bin` file and click **Install**.
4. Keep the device powered. The browser shows "Uploading..." and then "Installed. The device is restarting."

What you see on the device during an upload:

1. "Receiving firmware", with "From a browser" or "From PlatformIO".
2. A progress bar with "X of Y MB · N%" and the warning "Keep the device powered".
3. "Checking the firmware..."
4. "Installed · restarting...". The device restarts after about 1.5 seconds.

The running firmware is replaced only after the whole upload has been checked. With PlatformIO, `espot` prints "OK" before the device has checked the file. If the device refuses the file, the error appears only on the device.

While update mode is on

- The Settings rail icon shows an amber dot.
- The Wi-Fi icon in the status bar turns amber.
- Settings tiles show "Updating" or "Update mode".
- The screen stays lit.
- Incoming messages do not take over the screen.
- Opening another page on top (for example Wi-Fi settings) keeps update mode running.
- Turning Wi-Fi off from the quick panel shows a warning: "Update mode stops and the device restarts."
- You cannot start an online check or install. Trying gives "Update mode is on".

How update mode ends

Update mode always ends with a restart, which turns Bluetooth back on. The restart screen shows "Restarting...", the reason, and "Bluetooth comes back after the restart".

Cause	Message
You tap Stop , or leave the page with Back	"Update mode off"
10 minutes with no signed-in activity	"Update mode off (timed out)"

Cause	Message
30 minutes in total	"Update mode off (30-minute limit)"
10 wrong passwords	"Update mode off (wrong passwords)" (red)
Wi-Fi connection lost	"Update mode off (Wi-Fi lost)"
Upload sends no data for 20 s	The device closes the connection. If it is still stuck 15 s later, it restarts.

Errors

On the device, errors appear on a red screen with an **OK** button:

Device error	Meaning
"Not a firmware image" / "Image rejected" / "Image rejected (checksum or chip)"	The file is not valid firmware for this chip.
"Not a Wio L2 touch firmware"	The firmware is for a different build or board.
"File too big" / "Could not start (too big?)"	The file does not fit in the firmware slot.
"Filesystem images are blocked"	Only firmware files are accepted.
"Upload interrupted"	The transfer stopped before it finished.
"Could not reach the computer"	The device lost contact with the computer.
"Could not check the firmware" / "Update failed"	The final check or install step failed.
"Not enough memory (N KB free)"	The device does not have enough free memory to update.
"Refused, but could not undo: reflash over USB"	Flash the firmware over USB (see below).

Messages in the browser:

Browser message	Meaning
"Choose a .bin file first"	No file was selected.
"That is not a firmware file" / "File too big"	The file was refused.
"Password expired, reload"	The password changed or update mode restarted.
"Connection lost"	The browser lost contact with the device.
"Update mode is stopping"	Update mode is ending.
"Already installed, restarting"	An upload has already been installed.
"No file received"	The upload arrived empty.

Update over USB

USB flashing uses one of two image files:

File	Contents	Write at offset	Use it for
MeshCore-WioL2Pro-Cairn<N>-<date>-full.bin	Bootloader, partition table and app	0x0	A first flash, coming from Meshtastic, or recovery
MeshCore-WioL2Pro-Cairn<N>-<date>-update.bin	App only	0x10000	Updating an existing Cairn install

.uf2 files are for other boards. Do not use them on the L2.

To flash over USB:

1. Connect the device to the computer with USB.
2. To enter bootloader mode, hold **User/Boot**, tap **RST**, then release **User/Boot**.
3. Write the image with esptool: - Full image: `esptool.py --chip esp32s3 write_flash 0x0 ...-full.bin` - Update image: `esptool.py --chip esp32s3 write_flash 0x10000 ...-update.bin`
4. Press **RST** to start the new firmware.
5. Check that the slot chip reads "app0 · USB".

A browser flasher such as esptool-js also works. Use offset 0x0 for the full image.

Important: After a Wi-Fi update, flashing only the update image over USB can appear to work but change nothing. The device keeps starting from the other slot. Flash the `-full.bin` instead. Or write the blank boot-selection file too: `write_flash 0xe000 .../boot_app0.bin 0x10000 ...-update.bin`.

Returning to Seeed's factory firmware

Cairn replaces the firmware the board shipped with. To go back, flash Seeed's own firmware for the Wio Tracker L2 with Seeed's tools and instructions (see the Seeed wiki). A full flash replaces Cairn completely, including its contacts, chats and settings.

Maintenance

To open it, go to Settings → System and tap the **"Restart"** card. The page is titled **"Maintenance"**. You cannot reach it while the device is locked.

The card summary reads "Last: \". It turns amber after a crash, a watchdog reset or a brownout. The page shows "Last restart: \ · \ ago".

Option	How	What it does
Restart	Tap, then Restart in "Restart now?"	The mesh and Bluetooth drop for about 10 seconds. Nothing is lost. The screen shows "Restarting..."
Reset UI settings	Tap, then Reset in "Reset UI settings?"	Resets units, tones, volume, buttons, lock touches and the map view. Radio, contacts, chats, Wi-Fi, the time zone and "Keep across restarts" are kept. Toast: "UI settings reset".
Keep across restarts	Toggle	On: chats are saved. Toast: "Chats will be kept across restarts".
Hold to clear chat history	Hold 1.5 s, then Clear	Deletes every stored message on this device and the saved copy. Toast: "Chat history cleared".

Option	How	What it does
Hold to factory reset (red)	Hold 2 s, then Erase in "Factory reset?"	Erases contacts, channels, chats, this node's identity, Wi-Fi and every setting, then restarts. Maps on the card are kept.

If you turn **Keep across restarts** off, the device asks "Stop keeping chats?" first. Tap **Stop** to confirm. The saved copy is erased at once. Chats on screen stay until the next restart. The toast reads "Chats no longer kept".

The chat history status line can read:

- "Off · chats start empty after a restart"
- "N messages · saving shortly"
- "N messages saved X ago"
- "Couldn't save · trying again" (amber)

During a factory reset, the screen shows "Erasing everything..." / "Don't power off. This can take a minute; the device restarts when it is done." The erase usually takes 10 to 40 seconds. If it fails, the screen shows "Erase failed" / "Restarting..."

Last-restart reasons

About shows the full reason. The Settings card shows a short form.

About	Card	Meaning
"Power on"	"Power on"	Normal power-up.
"Restarted in Settings"	"Restart"	You tapped Restart .
"Factory reset"	"Erased"	After a factory reset.
"Firmware update"	"Update"	After a firmware install.
"Update mode ended"	"Update off"	Update mode stopped.
"Radio restore failed"	"Radio fault"	The radio settings could not be restored.
"Woke (low battery)"	"Low battery"	Powered on after an automatic low-battery shutdown.
"Woke from hibernate"	"Hibernate"	Powered on with User after hibernate.
"Reset pin"	"Reset pin"	RST was pressed.
"Crash"	"Crash" (amber)	The firmware crashed.
"Watchdog (stall)" / "Watchdog (interrupt)" / "Watchdog"	"Watchdog" (amber)	The device restarted itself after it stopped responding.
"Brownout (low power)"	"Brownout" (amber)	The supply voltage dropped too low.
"Software restart", "Woke from sleep", "SDIO reset", "Unknown"	e.g. "Sleep wake"	Other causes.

If the screen stops responding for 15 seconds, the device restarts itself. The reason then reads "Watchdog (stall)". About's "LAST RESTART" card also shows "Watchdog 15s", "Watchdog paused" or "Watchdog off".

Rescue CLI (advanced)

The rescue CLI lets you repair the device's storage over a USB serial connection. Use it only if the normal interface cannot fix the problem.

To enter the rescue CLI:

1. Connect the device to a computer over USB and open a serial terminal.
2. Restart the device.
3. While the boot splash is showing (about 3 seconds), long-press **User** for 1 second.
4. The toast reads "Rescue CLI (USB) · restart to leave".

A long press after the splash runs your normal Long-press action instead. If the device is locked, a long press shows the lock hint and does not start rescue.

While the rescue CLI is active, the app link and saving are stopped, and the watchdog is paused. About shows "Watchdog paused". Restart the device to leave the CLI.

Command	What it does
<code>set pin <0 or 6 digits></code>	Sets the Bluetooth PIN.
<code>erase</code>	Formats storage.
<code>rebuild</code>	Formats storage, then saves the identity, preferences, contacts and channels again.
<code>ls <path></code>	Lists files.
<code>cat <path></code>	Shows a file.
<code>rm <path></code>	Deletes a file.
<code>reboot</code>	Restarts the device.

Troubleshooting

Symptom	Fix
"Start update mode" is missing; the button reads "Open Wi-Fi settings"	Wi-Fi is not connected. Connect to a network, then return to the page.
Phone app lost Bluetooth during or after an update	This is expected. Bluetooth comes back after the automatic restart.
Browser shows "Password expired, reload"	Reload the page and sign in with the password now on the device.
PlatformIO said "OK" but the build did not change	The device refused the file. Read the error on the device.
"Not a Wio L2 touch firmware"	Use a Cairn file for the Wio L2 touch build.
"Refused, but could not undo: reflash over USB"	Flash the <code>-full.bin</code> over USB at 0x0.
Update mode ended on its own	Check the reason on the restart screen. Start update mode again, then upload within 10 minutes.
A USB flash "worked" but the build did not change	The device is starting from the other slot. Flash the <code>-full.bin</code> , or also write <code>boot_app0.bin</code> at 0xe000.
Online install refused with "Update mode is on"	Stop update mode first.
"Already busy"	A check or install is already running. Wait for it to finish.

Symptom	Fix
"Check for updates first"	Check for updates, then tap Install again.
"Connect to Wi-Fi first"	Connect to Wi-Fi, then tap the online-check card again.
"Certificate not trusted"	GitHub's certificate changed. Install a newer build over USB or with update mode.
"GitHub busy · try again later"	GitHub limits requests to 60 an hour for each address. Try again later.
"Update failed · \" after an online install	The device is still on the old firmware. Try again, or update from a computer.
Last restart shows amber "Crash", "Watchdog" or "Brownout"	The device restarted because of a fault. "Brownout" means the power dropped too low.
Factory reset seems stuck on "Erasing everything..."	Wait. It can take up to about a minute. Do not power off.
Rescue CLI did not start	Long-press User while the boot splash is still on screen, and make sure the device is unlocked.

12 Reference

This chapter collects the buttons, terms, hardware facts and credits you may want to look up quickly.

Button cheat sheet

The User button has three actions you can change. Set them in **Settings** → **Controls** → **Buttons**.

Button	Press	Action
User	Click (press once)	Quick panel (default, configurable). The action starts after a short pause while the device waits for a second press.
User	Double-click (press twice)	New message (default, configurable). Unlocks the device when it is locked.
User	Long press (hold 1 s)	Back (default, configurable). On a tab's first page, Back turns the screen off.
User	Triple-click	Mute or unmute ("Sound muted" / "Sound on"). This cannot be changed. Unmuting plays a confirmation tone.
User	Long press during the boot splash (about the first 3 s)	Starts the USB rescue CLI ("Rescue CLI (USB) · restart to leave"). The app link and saving stop until you restart.
User	Any press while hibernating	Powers the device back on
WAKE	Click, screen on and unlocked	Turns the screen off. If "Auto lock" is on, the device locks too.
WAKE	Click, locked and lock screen lit	Turns the screen off
WAKE	Hold 1 s, screen on and unlocked	Locks now, with a tone and the toast "Locked - 2x WAKE". The screen goes dark 1.5 s later.
WAKE	Hold 1 s, locked and lock screen lit	Nothing
WAKE	Click or hold, screen dark	Wakes the screen. If the device is locked, the lock screen lights up.
WAKE	Double- or triple-press, locked	Unlocks, even from a dark screen
WAKE	Double- or triple-press, unlocked and lit	Nothing
RST	Press	Hardware reset. Afterwards, About shows "Reset pin" as the last restart.
User + RST	Hold User, tap RST, release User	Starts the ESP32-S3 flasher for USB flashing
Power switch	Slide	Hardware power on or off. Prefer Hibernate for a clean shutdown.

You can assign any of these actions to the User button's click, double-click and long press:

Choice	What it does
"None"	Nothing. It also does not close an open panel, sheet or dialog.
"Quick panel"	Opens the Quick panel. A second press closes it.
"Back"	Closes an open sheet or goes back one page. On a tab's first page, it turns the screen off.

Choice	What it does
"New message"	Opens Chats with the recipient picker
"Send advert"	Opens the Zero-hop / Flood advert sheet. Nothing is sent until you tap Send.
"Ping last node"	Asks "Ping \?" for the node you last heard from or messaged
"Scan repeaters"	Opens the Repeater scan and starts a scan. During the cooldown it shows "Scan again in Ns".
"Toggle GPS"	Off → On, then On ↔ Eco
"Flashlight"	Turns the whole screen white at 100 % brightness. A touch or a User press ends it. A WAKE click ends it and turns the screen off.

"New message" and "Scan repeaters" ask "Discard changes?" first if you have unsaved edits.

These User button rules always come first, in this order:

1. **Locked:** a click or long press only lights the lock screen for about 2.5 s. A double- or triple-press unlocks.
2. **Screen dark:** any press only wakes the screen.
3. **Flashlight on:** any press only turns the flashlight off.
4. **Triple-click:** always mutes or unmutes.
5. **Panel, sheet or dialog open:** a press closes it instead of running its action. This does not apply if the action is set to "None".

WAKE cannot power the device back on from hibernate. Only User can.

Glossary

Term	Meaning
Advert	A short broadcast that announces your node on the mesh. It carries your node name, and your position if "Share position in adverts" is on. With position, the name is limited to 23 bytes. Without it, the limit is 31 bytes.
Repeater	A node that relays other nodes' packets so they travel farther. You can find nearby repeaters with Repeater scan .
Room server	A mesh node that hosts a shared message room. You log in to it, and it keeps posts for the people who join.
Channel	A group conversation that everyone with the channel's key can read. Cairn has three kinds: Public, Hashtag (the key comes from the name) and Private (the key is shared with members).
Hop	One relay step. A packet that a repeater passes on has taken one more hop.
Flood / direct	A flood packet is relayed across the whole mesh. A direct packet follows a known path. A zero-hop advert reaches only the nodes in direct range.
SNR	Signal-to-noise ratio, in dB. It shows how far a received signal stands above the background noise. Higher is better.
Noise floor	The background radio noise level, in dBm. The Quick panel shows it as "NF". "NF --" means it has not been measured yet.
Airtime	How long a transmission occupies the radio channel. The Radio settings show it as "Airtime \~N ms per 40-byte message".

Hardware summary

These are the specifications Seeed publishes for the Wio Tracker L2 Pro.

Part	Specification
Processor	ESP32-S3
LoRa radio	Semtech SX1262, 863–928 MHz
Display	3.2" IPS touchscreen, 320 × 240
GNSS	Quectel L76K
Audio	ES8311 codec, 6 Ω 0.8 W speaker, microphone
Battery	3000 mAh
Storage	microSD card slot
Expansion	Grove connector
Power and data	USB-C
Buttons and switches	User (Boot), WAKE, RST, power switch

In Cairn, transmit power ranges from –9 to 22 dBm. Keep it within your region's limit and Seeed's rating for the board (20 dBm on US915).

Credits and licence

- Cairn is a fork of MeshCore. MeshCore is released under the MIT licence, Copyright (c) 2025 Scott Powell / rippleradios.com.
- Seeed Studio and Wio Tracker are products of Seeed Studio.
- Cairn is an independent project. It is not affiliated with Seeed Studio.
- Cairn firmware, this manual and the case design: Christopher Van Hoose · 